TianshanOS

▽ The device management system for RM-01

API Integration Guide

Complete API Reference

Getting started, roles, limits and result checks

HTTP / WebSocket / SSH / Configuration / OTA

Version covered: TianshanOS 0.6.4

English edition

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This guide can still be used with later versions where interfaces and procedures remain unchanged. If parameters, messages, or results differ, check the documentation for your installed version.

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TianshanOS API User Guide

Connect your own scripts and systems to monitor the device, control its computers, receive live status, run remote tasks, and maintain configuration and updates. This guide assumes familiarity with HTTP, JSON, and basic server operations; circuit and firmware knowledge is not required.

This guide is based on **TianshanOS 0.6.4**.

Find your task

Task	Start here
Make your first API call	Quick start
Identify the controller and the two computers	Devices and names
Monitor temperatures, fans, power, and computers	Monitoring
Switch power or the USB connection	Device control
Start and verify a remote service	Remote tasks
Configure variables, sources, actions, and rules	Automation and Configuration models
Transfer files or update firmware	Maintenance and Transfers
Receive live status and results	WebSocket
Look up an API	Complete reference
Investigate errors or maintain a long-running integration	Troubleshooting and Operations

Reading order

Complete a read-only request in the quick start first. Then follow the scenario for your task, including its result checks and recovery steps. Open the reference when you need individual parameters.

The documented default accounts admin/root and initial password rm01 are product defaults. Other addresses, filenames, tokens, and response values are illustrative. Use the actual address and current password of a device you manage. Examples contain no private tokens, changed passwords, or real application records.

The reference covers 277 registered names in 28 functional groups, including config.pack.apply, which is retained but does not apply generic configuration. Dedicated file/image routes and WebSocket messages have separate chapters. A registered name does not guarantee that the required hardware, storage, credentials, or remote service is ready.

Quick start

Connect, log in, and read a protected measurement. Apart from creating and closing a login session, this sequence only reads status.

Set the device address

Use the device's current IP address. WebUI and business APIs use HTTP port 80 by default, so no explicit port is normally needed. The API prefix is `/api/v1`. Replace the example address with your own device address:

SH Source: example-001.sh

```
export TS_BASE='http://10.10.99.97'
```

For connection problems, check the IP address, network, and service state. See Troubleshooting.

Read controller information

SH Source: example-002.sh

```
curl --connect-timeout 5 --max-time 15 \
  "$TS_BASE/api/v1/system/info"
```

Illustrative response excerpt:

JSON Source: example-003.json

```
{
  "code": 0,
  "data": {
    "app": {
      "name": "TianShanOS",
      "version": "0.6.4",
      "compile_date": "Nov 5 2026",
      "compile_time": "10:00:00"
    },
    "chip": {
      "model": "ESP32-S3",
      "cores": 2
    },
    "uptime_ms": 123456
  }
}
```

Check HTTP status and then code. Version is at data.app.version; uptime_ms is milliseconds. Do not assume a top-level data.version field.

Log in and retain the session

The default accounts are `admin` and `root`, both with initial password `rm01`. Log in and use the APIs permitted for that account. If you have changed the password, use your current password.

Set `umask 077` before creating login.json and restrict access to the file. This example uses the default admin account:

JSON Source: example-004.json

```
{"username":"admin","password":"rm01"}
```

SH Source: example-005.sh

```
umask 077
curl --connect-timeout 5 --max-time 15 \
  -H 'Content-Type: application/json' --data-binary @login.json \
  "$TS_BASE/api/v1/auth/login" > login-response.json
export TS_TOKEN="$(python3 - <<'PY'
import json
r = json.load(open("login-response.json"))
assert r["code"] == 0
print(r["data"]["token"])
PY
)"
```

Successful data contains token, session_id, username, level, expires_in in seconds, and password_changed. Use token as the credential; session_id is only an identifier.

Use the returned token for subsequent calls. Access follows data.level and endpoint authorization. Password changes are an optional account-management operation through auth.change_password.

Handle password files, login responses, and tokens as credentials. Remove temporary credential files after use.

Read a protected temperature

SH Source: example-006.sh

```
curl --connect-timeout 5 --max-time 15 \
  -H "Authorization: Bearer $TS_TOKEN" \
  "$TS_BASE/api/v1/temp/read"
```

Illustrative excerpt:

JSON Source: example-007.json

```
{
  "code": 0,
  "data": {
    "temperature_c": 42.5,
    "temperature_01c": 425,
    "source": "variable",
    "timestamp_ms": 120000,
    "valid": true
  }
}
```

temperature_c is Celsius; temperature_01c is tenths of a degree. Only use the reading as valid when valid:true. Missing or invalid data must not become 0 degrees in your monitoring system.

Log out

SH Source: example-008.sh

```
curl --connect-timeout 5 --max-time 15 \
  -X POST -H "Authorization: Bearer $TS_TOKEN" \
  "$TS_BASE/api/v1/auth/logout"
unset TS_TOKEN
```

POST logout returns `{"success":true}`, rather than the usual code/data envelope. Scripts and connections using the session need a valid session again.

Next, choose a task scenario or use the read-only Python example supplied with the editable documentation.

Devices and names

TianshanOS manages both its controller and the connected computers. Identify the target before calling an API: their status values describe different things.

User-facing name	API identifier	Purpose
Management controller	system.*	TianshanOS version, memory, tasks, and controller restart
Inference computer	device:"agx"	Power state, power control, and reset; monitor.* provides compute telemetry
Application computer	device:"lpmu"	Power state, power control, reset, and upper-level network access setup
Cooling fans	fan.* and numeric id	RPM, control mode, temperature curve, and output limits

User-facing name	API identifier	Purpose
Touch light, board light, matrix display	touch, board, matrix	Colors, effects, images, QR codes, and text
Local storage	/sdcard/... and other supported paths	Files, configuration packages, and supported update materials
Remote host	Saved SSH host id/host	Server commands or managed processes; not necessarily one of the built-in computers

Use agx and lpmu exactly as the API identifiers. For example, the application computer is still addressed as `{"device":"lpmu"}`.

Interpret measurements correctly

- **Fan duty** is the output percentage, from 0 to 100. RPM is rotational speed. Unknown duty may be null; also check duty_valid.
- **Temperature source** may be a local sensor, compute telemetry, a variable, or a manual test value. Check the source, valid flag, and timestamp.
- **power_good** is a hardware power indication. It does not establish that the operating system or application is ready.
- **USB target** selects a shared USB route. Switching it can interrupt traffic on the previous route.
- **run_id** identifies a managed remote service run. A retained record does not establish that its process is still running; verify its status.
- **Operation receipt** contains an operation identifier and result credentials. Preserve the complete receipt rather than just a display name.

Three stages of completion

1. The API accepts the request, for example by returning started:true.
2. The device action or remote process reaches the requested state.
3. Your application becomes ready, for example by passing a health check.

Confirm the stage your task requires. Restarting the controller, powering a computer, and starting its application are separate operations.

Task scenarios

Examples use TS_BASE and TS_TOKEN from the quick start. Replace addresses, object IDs, configuration versions, and receipts with actual values. Control examples have real effects: use only the task you intend to perform.

Monitor the device

Read temperatures, cooling, power, and computer status into your monitoring system. Log in for READ capabilities; some overview endpoints are public.

SH Source: example-009.sh

```
curl -H "Authorization: Bearer $TS_TOKEN" "$TS_BASE/api/v1/temp/read"
curl -H "Authorization: Bearer $TS_TOKEN" "$TS_BASE/api/v1/fan/status"
curl -H "Authorization: Bearer $TS_TOKEN" "$TS_BASE/api/v1/power/status"
curl -H "Authorization: Bearer $TS_TOKEN" "$TS_BASE/api/v1/device/status?device=agx"
```

Retain temperature valid, fan duty_valid, power valid/crc_valid, and last update times. Display invalid or disconnected states explicitly. duty is a percentage; rpm is rotational speed. Computer state=on does not establish application readiness. See Response models.

Successful monitoring means valid readings that keep updating. Use WebSocket for a continuous dashboard. For invalid temperatures, inspect temp.sources; for compute telemetry problems, inspect monitor.status. Do not replace missing values with zero. The supplied Python example performs login, reads, and logout only.

Control a computer

ADMIN is required. First read device.status to identify agx or lpmu and its current state. Use an explicit on/off action rather than repeating toggle after a timeout.

SH Source: example-010.sh

```
curl -H "Authorization: Bearer $TS_TOKEN" \
  -H 'Content-Type: application/json' \
  --data-binary '{"device":"agx","action":"on"}' \
  "$TS_BASE/api/v1/device/power"
```

Then GET device.status and inspect state/power_good. Confirm application readiness on the target computer or through a business health endpoint. For normal shutdown use off. force_off/reset interrupt work and can lose data; arrange target-system maintenance first.

To switch USB, POST device.usb.set with `{"target":"agx"}`, then GET device.usb.status. Other targets are esp32/lpmu/disconnect. The old route may disconnect during the switch; that alone is not a device fault.

After a timeout, query state before retrying. Preserve error state, power indication, and details if the target does not reach the expected state.

Adjust lights and fans

Routine LED operations need ADMIN. GET led.list to obtain a device name, then fill it:

SH Source: example-011.sh

```
curl -H "Authorization: Bearer $TS_TOKEN" \
  -H 'Content-Type: application/json' \
  --data-binary '{"device":"board","color":"#00A0FF"}' \
  "$TS_BASE/api/v1/led/fill"
```

Read led.list again. To use the display after restart, separately POST led.save with `{"device":"board"}` and inspect saved. Displaying and saving startup settings are different steps.

For fans, read fan.status/config first. POST fan.set with `{"id":0,"duty":50}` only after obtaining the actual fan ID. fan.mode selects off/manual/auto/curve. Curve, limits, and persistence permissions are listed in the Fan reference.

Check duty_valid/duty/target_duty, temperature, and faults after changing output. Restore the intended control mode after a temporary manual setting.

Manage remote tasks

Use ssh.exec/exec_stream for a one-off command; use a saved SSH command and run record for a continuously managed service. Raw command execution requires ROOT; starting an existing managed service requires at least ADMIN.

Before starting, confirm reachability, a verified host fingerprint, and a saved host referencing a valid SSH key. Unknown/changed fingerprints are verified through ssh.test and trusted information, not by clearing trust or automatically accepting.

1. With root, read ssh.hosts and ssh.commands. Check host, command, service_mode, and readiness matching. See Configuration models.
2. Read the latest active_version and start the saved service:

SH Source: example-012.sh

```
curl -H "Authorization: Bearer $TS_TOKEN" \
  -H 'Content-Type: application/json' \
  --data-binary \
  '{"command_id":"YOUR_SERVICE_COMMAND_ID","active_version":"CURRENT_VERSION"}' \
  "$TS_BASE/api/v1/ssh/services/start"
```

3. Retain run_id and operation_id. POST automation.services.status with `{"run_id":"YOUR_RUN_ID","verify":true}`. Inspect operation_phase, busy, and state. verify connects to the original host; it is not just a local cache read.
4. Verify application health/readiness. To stop, POST automation.services.stop with the original run_id and continue querying status.

A run_id is not a ready indication; last_known is a previous observation. Deleting configuration or a local key does not stop remote processes. Use the original run identity rather than guessing a process by name.

For a one-off stream, authenticate WebSocket before POST ssh.exec_stream. Keep the complete receipt and use ssh.exec_status after interruption. Never rerun a command merely to retrieve missing output. See Streaming SSH for execution, matching, and output completeness.

Configure automation

The flow is: acquire data, map it into variables, then execute an action when conditions hold. With root, read automation.status/config.status first. Resolve loading, enablement, pause, or SD-source selection issues before editing.

Start with a saved user variable:

SH Source: example-013.sh

```
curl -H "Authorization: Bearer $TS_TOKEN" \
  "$TS_BASE/api/v1/automation/config/status"
```

Use its active_version in the next request:

SH Source: example-014.sh

```
curl -H "Authorization: Bearer $TS_TOKEN" \
  -H 'Content-Type: application/json' \
  --data-binary \
  '{"name":"example.temperature","value":42.5,"active_version":"CURRENT_VERSION"}' \
  "$TS_BASE/api/v1/automation/variables/set"
```

Read automation.variables.get with root to confirm value, readonly, and source_id. Retain the new active_version. Create an enabled:false rule from the rule model; use a log action before adding power or remote-execution effects.

Add a real data source only when needed. Verify data arrival, variable updates, and stale status before enabling the rule. Saving a definition, enabling configuration, starting the engine, and executing actions are separate states; saved:true alone is insufficient.

Refresh and merge after a conflict. Preview deletion impact. Preview/apply SD or backup selection. Respect pause/busy results rather than overwriting managed files.

Maintain the device

Choose one maintenance goal at a time so its effects can be checked.

Goal	Sequence	Completion
Back up an object	Export with the role required by that object; save tscfg; preview any later import	Complete package retained and verifiable; export is not recovery
Save a file	Upload, inspect attributes, download and compare	Intended path, length, and bytes

Goal	Sequence	Completion
Update firmware/WWW	Check matching images, transfer, read global progress, confirm state, restart if needed	Correct running identity/version and working application
Change networking	Record old access/configuration, change, save, reconnect	Management and target networking both work
Change a password	change_password, log in again, update client sessions	New credentials work; old sessions are no longer assumed valid
Install identity certificate	Check CSR/key match, install, inspect cert.status and client trust	Correct material, actual loaded identity, and verified client connection

See Transfers and updates for complete requests and capacity limits. cert.* manages device identity, key.* manages SSH keys, and hosts.* manages remote fingerprints. After failure, establish which stages took effect before retrying or recovering.

Common calling rules

Paths and methods

Business URLs have the form `http://10.10.99.97/api/v1/...`. Dots in a logical name become path separators: fan.status maps to /api/v1/fan/status. Preserve case, including key.exportPrivate and log.setLevel.

Use only the methods listed for each API. Read operations usually allow GET/POST; writes usually allow POST/PUT/DELETE, with some POST-only endpoints. This guide generally uses GET for simple queries and POST for actions. A disallowed method returns HTTP405.

Names and parameters determine the action. Do not infer that DELETE always deletes an object or POST always creates one.

Sessions and roles

Protected requests carry `Authorization: Bearer <token>`. Do not place tokens in URLs. A role or session_id supplied in ordinary request JSON does not establish that identity.

Access label	Meaning
Public	This capability can be accessed without login; resource conditions still apply
READ	At least read permission; authorized admin/root sessions normally qualify
ADMIN	Routine device operations; admin or root required

Access label	Meaning
ROOT	Raw command execution, automation definition writes, and selected account management; root required
Body token	Authentication APIs validate the token JSON field; this is not anonymous permission to change accounts

Default accounts are admin/root, both with initial password rm01. The policy distinguishes READ/WRITE/ADMIN/ROOT, but this does not imply an API for creating arbitrary accounts at all levels. No registered endpoint has minimum WRITE in this version.

Some resources or parameters add checks: ordinary file reads may allow READ; sensitive paths require ADMIN. config.get exposes only the language preference publicly; other keys require ADMIN. Full SSH host/command, rule, and action reads normally require ADMIN; secrets in automation.sources.list still require ROOT. temp.bind queries allow ADMIN, while binding changes require ROOT. automation.proxy.fetch inherits destination permissions for local APIs and requires ROOT for external services. restricted:true means fields were filtered; never save omitted fields back as empty values.

Parameters

Use Content-Type: application/json and a JSON object body. Arrays belong inside object fields. Avoid duplicate keys and overlapping query/body parameter names; conflicts are rejected.

Query values are decoded once and may be interpreted as integers or booleans. Prefer JSON for objects, arrays, floating-point values, null, and identifiers whose leading zeroes must be preserved. Do not URL-encode twice.

Optional means that omission is permitted; null is not generally equivalent to omission. Some binding APIs explicitly use null to clear a binding. Time units are field-specific: milliseconds, seconds, or Unix time.

Determine the result

Usual success:

JSON Source: example-015.json

```
{"code":0,"data":{"example":"value"}}
```

Business errors can still use HTTP200:

JSON Source: example-016.json

```
{"code":1,"error":"invalid_request"}
```

Explanations may be in error or message. data may be an object, an array, or absent. A successful response need not contain message.

1. Handle non-success HTTP status as a transport, authorization, format, or service failure.
2. If code is present and nonzero, handle a business error.

3. Check the endpoint's valid, success, exit_code, persisted, or operation status. A top-level success does not replace these checks.
4. For asynchronous work, use the completion flow specified by that API; retain a receipt when one is returned. After a disconnect or result_unknown, inspect state before retrying.

Login, logout, binary uploads, and downloads have dedicated formats. A successful download body contains file bytes, not JSON.

Error codes

code	Name	Suggested response
0	OK	Check the endpoint's completion conditions
1	INVALID_ARG	Check fields, types, ranges, and version conditions
2	NOT_FOUND	Refresh object/operation information; do not invent or reuse stale IDs
3	NO_PERMISSION	Check role, sensitive resources, and export conditions
4	BUSY	Inspect active work and wait before retrying
5	TIMEOUT	Determine whether a result exists before repeating a write
6	NO_MEM	Reduce request size or load; preserve known results
7	INTERNAL	Retain the reason; investigate state if result_unknown
8	NOT_SUPPORTED	Use a supported entry point
9	HARDWARE	Inspect the device, sensor, and configuration
10	CONNECTION	Check network, remote host, fingerprint, and service
11	AUTH	Reauthenticate or complete the required verification

SSH trust flows also use1001/1002 for unknown/changed host fingerprints and provide fingerprint details. Verify through a trusted channel before confirming; deleting trust records is not identity verification.

Effects and persistence

applied reports an applied setting; persisted or saved reports storage success. They can differ. An error may occur after an earlier stage has taken effect: retain its details and read back state.

Toggles, rule triggers, remote execution, and updates are not guaranteed safe to repeat. Read-only queries may be retried with appropriate load control. After a write timeout, query state or the receipt first.

Limits

Connections, bodies, receive times, and task slots are bounded. Identity and size checks can reject a request before consuming its entire body. Avoid replacing live subscriptions with large numbers of concurrent polls.

This version allows up to 1 MiB for the generic JSON route with a30-second receive budget; authentication requests are limited to 16 KiB/10 seconds. Configuration packages themselves are limited to 64 KiB. Individual APIs can impose smaller limits. See Transfers for binary routes.

JSON nesting is limited to 32. Embedded NUL, trailing garbage, and duplicate keys are rejected. For HTTP413, reduce the request or use the feature's designated upload route.

Common response models

These fields are inside data. Handle HTTP status and code as described in Common rules. Fields vary with device state, initialization, and role. Preserve unknown when a field is missing; do not fill it with0 orfalse.

temperature

Field	Interpretation
temperature_c / temperature_01c	Celsius / tenths of a degree; the same measurement
valid	Whether the value is usable; false must not be treated as normal temperature
source / active_source / preferred_source	This reading's source / actual source / preferred source, depending on the API
timestamp_ms / last_update_ms	Milliseconds since device start, not Unix time
stale / age_ms	Staleness / age in milliseconds; null may mean never updated
partial_stale / valid_count / stale_count	Partial failure of a multi-variable binding; inspect more than its average

The implementation bounds valid temperature values. Inputs can come from local sensors, compute telemetry, variables, or manual tests. A manual value does not establish the physical temperature.

fan

fan.status with id returns one fan; without id it returns fans plus temperature/temp_valid/temp_source.

Field	Type, unit, or meaning
id / mode	Fan number / off, manual, auto, curve
duty_valid / duty	Output validity / known actual duty in percent; unknown duty is null
target_duty	Target output percentage; can differ from actual duty
rpm	Revolutions per minute; interpret with tachometer capability and faults

Field	Type, unit, or meaning
enabled / running / fault	Enabled / running / fault; unknown running can be null
temperature	Relevant control temperature, Celsius
control_temperature / guard_temperature / predicted_temperature	Control, protection, and predicted temperature, Celsius
slope_c_per_min	Temperature slope in Celsius per minute
auto_state / guard_active / temp_stale	Automatic stage, active guard, stale temperature
controller_gain / cooling_response	Diagnostic values, not RPM or temperatures

applied and persisted distinguish application from storage. Requesting50% does not establish a valid physical50% output.

power

power.status combines voltage, chip measurements, protection, and statistics. Use the specific voltage/chip/protection.status APIs for details.

Field	Unit or meaning
voltage_v / current_a / power_w	V / A / W
voltage_mv	mV; convert before comparing to V
valid / crc_valid	Data validity / communication integrity; check both for chip measurements
timestamp_ms	Milliseconds since start
state / countdown_remaining_sec	Protection stage / countdown seconds
initialized / running	Initialization complete / protection running

device.power.status instead reports rails with voltage_mv/current_ma/power_mw and total_power_mw. Compute telemetry can also use mW; follow each field's unit.

config-result

Configuration writes and source selection return a set of state flags; not every response contains all of them.

Field	Meaning
active_version	Current configuration version; use the new returned value for the next write
source	Active configuration source

Field	Meaning
loaded / configured / enabled / paused	Loaded / configured / enabled / paused; separate states
saved / uncertain	Storage completed / outcome uncertain
load_error / internal_error	Active configuration or internal-backup error
sd_available / internal_exists / internal_valid	SD available / internal backup exists / backup valid
comparison_known / backup_matches	Comparison possible / matching backup; unknown is not a match
awaiting_sd_choice	An SD selection is required
internal_saved_at	Internal backup save-time information
collection / id / dependency	Collection, object, or dependency location
impact	References and runs affected by deletion/replacement

Preview package_digest and credential_generation are apply preconditions, not reusable global authorization. restart_required:false does not replace loaded/enabled or runtime checks.

variable

Variables include name, type/value_type, and possibly value, source_id, persistent, readonly, flags, last_change_ms, last_update_ms, age_ms, and stale. variables.list uses include_value for values and include_meta for metadata. A missing value field is not a null value.

certificate

cert.status separately reports has_private_key, has_certificate, has_ca_chain, certificate information, and the separate HTTPS diagnostic-service status. Certificate validity may depend on time_ready; undetermined is not valid.

An installed identity, the identity loaded by the separate HTTPS diagnostic service, and diagnostic-client trust are separate checks. Inspect service status after installation. Business APIs remain on HTTP; their Bearer roles are separate from mTLS certificate roles.

color

RGB components are0-255. Direct color APIs accept supported strings, r/g/b objects, or the decimal integer equivalent of 0xRRGGBB; JSON does not allow hexadecimal numeric literals. led.color.hsv uses h=0-359 degrees and s/v=0-255, not percentages.

Color correction is an object: enabled; white_point with enabled/red_scale/green_scale/blue_scale; gamma with enabled/gamma; brightness/saturation with enabled/factor. Scales range0-2; gamma ranges0.1-4. Send nested objects when modifying them. Device brightness0-255 and a correction factor are different units.

ota

After a URL or SD start, read ota.progress. It reports device-wide state, without a private per-login receipt. APP progress has state/percent/total_size/received_size/message; WWW progress has state/percent/total/received/message/running. APP pending_reboot means a restart is needed; WWW completion uses state=completed and running=false. 100 percent alone does not prove the new version is running.

Partitions report label/address/size, bootability, and available version/project. can_rollback reports current rollback availability. aborted acknowledges the cancel call; continue checking progress and actual state.

Binary uploads use a separate response. HTTP409 with status=partial, error=recovery_cache_failed, and flash_written=true means Flash was written but the SD recovery cache was not fully saved. Inspect partitions and recovery files before retrying.

network

Ethernet interface_present reports interface existence; link_up reports the physical link. config_error/config_error_name report configuration-application errors. A connected cable does not establish a successful IP configuration.

DHCP running is actual state; Ethernet want_running is requested state. restore_error/apply_error/resume_error distinguish configuration restore, application, and service resumption failures. Zero means that step reported no error. Check interface and link state before deciding whether the service should run.

Configuration models

These models save hosts, commands, sources, actions, and rules that the device can use at runtime. Save a disabled example and read it back before enabling effects.

Read the version before writing

With root, read automation.config.status and retain active_version. Put it at the top level of a write. After each successful write, use the new version for the next request. For rule updates, also carry the read revision as expected_revision. A conflict requires rereading and merging, not merely replacing the version token and replaying stale input.

The JSON objects below are complete request-shape examples. Replace CURRENT_VERSION, identifiers, addresses, keys, and variable names. Place model fields at the request top level; do not add a config wrapper except for explicitly shown subobjects.

restricted:true can indicate a metadata-only view. Use ADMIN for complete SSH host/command, rule, and action reads; automation.sources.list secrets still require ROOT. Writing definitions follows each endpoint's required role. Omitted secret fields are not empty values. Host/source updates merge with existing JSON; command/action updates should carry complete definitions. Rules use their own merge/revision checks; do not generalize one family's update behavior.

SSH hosts

For ssh.hosts.add/update. This records where/how to connect and which key to use; hosts.* stores fingerprints instead.

JSON Source: example-017.json

```
{
  "id": "example-host",
  "active_version": "CURRENT_VERSION",
  "host": "192.0.2.20",
  "port": 22,
  "username": "YOUR_USER",
  "auth_type": "key",
  "keyid": "example-key",
  "enabled": true
}
```

Field	Type, limit, and meaning
id	Stable string, at most 31 bytes; must not already exist for add; must exist for update
host	Address string, at most 63 bytes
port	SSH integer port, normally22; use1-65535
username	Remote user, at most 31 bytes; distinct from device login
auth_type	key/password; key is recommended for persistent definitions
keyid	Saved local SSH private-key ID, at most 31 bytes
enabled	Boolean, decoding default true
created_time	Optional unsigned 32-bit time value; normally retain the existing value

The persistent model does not store a password field. password mode does not imply a saved remote password; immediate calls supply their own credentials. At most16 hosts. Address, key existence, and remote trust are checked separately.

SSH commands

For ssh.commands.add/update. Saving does not execute a command. Use service_mode:false for short commands; managed long-running services use true and ssh.services.start.

JSON Source: example-018.json

```
{
  "id": "example-command",
  "active_version": "CURRENT_VERSION",
  "host_id": "example-host",
  "name": "Read OS",
  "command": "uname -a",
  "timeout_sec": 30,
  "enabled": true,
  "service_mode": false
}
```

Field	Type, limit, and meaning
id / host_id	Strings, at most 31 bytes each; host_id references a saved host
name / command	Strings, at most63 /1023 bytes
desc / icon	Optional strings, at most127 /63 bytes; icon can be a display icon or path
expect_pattern / fail_pattern / extract_pattern	Optional match/extraction patterns, at most 127 bytes each
var_name	Optional result-variable prefix, at most 31 bytes
timeout_sec	0-65535 seconds, default 30; not milliseconds
stop_on_match / nohup / enabled / service_mode	Booleans; enabled defaults true, others false
ready_pattern / service_fail_pattern	Service readiness/failure patterns, at most 127 bytes each
ready_timeout_sec / ready_check_interval_ms	Integers0-65535, seconds/milliseconds respectively; configure deliberately for services
created_time	Optional unsigned 32-bit time value

Aliases are timeout, expectPattern, failPattern, extractPattern, varName, stopOnMatch, serviceMode, readyPattern, serviceFailPattern, readyTimeout, readyInterval. Prefer table names; avoid supplying both aliases. _object_id/runtime_instance represent persistent/runtime identity; retain returned values rather than inventing them. At most64 commands.

Data sources

For automation.sources.add/update. Sources acquire data and map it to variables used by rules.

JSON Source: example-019.json

```
{
  "id": "example-source",
  "active_version": "CURRENT_VERSION",
  "label": "Temperature",
  "type": "rest",
  "enabled": false,
  "poll_interval_ms": 5000,
  "auto_discover": false,
  "rest_url": "https://example.com/status",
  "rest_method": "GET",
  "mappings": [
    {
      "path": "temperature",
      "var": "compute.temperature"
    }
  ]
}
```

Shared field	Type and meaning
id / label	Strings, at most63 /47 bytes
type	builtin/websocket/socketio/rest/variable strings, or corresponding integers0-4
enabled / auto_discover	Booleans; auto_discover defaults true; example explicitly disables it
poll_interval_ms	Integer0-4294967295 milliseconds; update behavior depends on type; zero is not unlimited sampling
mappings	At most4: path up to 95 bytes, var up to 63 bytes, optional transform up to 63 bytes

Mapping path/var also accept json_path/var_name. The path locates a field in data, such as temperature or cpu.cores[0].freq; it is not a URL. Verify transforms with known sample values.

Type	Type-specific fields
websocket	ws_uri up to 95 bytes; ws_path legacy single-field path up to 63 bytes; reconnect_ms0-65535 ms
socketio	sio_url up to 95 bytes; sio_event up to 63 bytes
rest	rest_url up to 95 bytes; rest_method up to 7 bytes; rest_auth up to 47 bytes; rest_path legacy path up to 63 bytes
variable	var_name, var_prefix, var_watch_all; SSH data flows also use ssh_host_id, ssh_command, expect_pattern, fail_pattern, extract_pattern, timeout_sec
builtin	builtin object: type selects a built-in sample; GPIO uses pin, ADC uses channel/atten, computer state can use device_index

String-type inputs also accept convenience aliases: WebSocket uri/json_path; Socket.IO url/event; REST url/method/auth_header/json_path. Prefer canonical fields above. Variable-source strings are generally at

most 63 bytes, ssh_command 95 bytes, timeout_sec0-65535 seconds. For built-in numeric hardware type/channel IDs, start from existing definitions rather than guessing pins or channels.

User variables

automation.variables.set writes; get/list reads. Name, source, readonly, and age matter in addition to value.

JSON Source: example-020.json

```
{"name":"example.temperature","value":42.5,"active_version":"CURRENT_VERSION"}
```

value supports null, boolean, finite number, or a string up to 63 bytes. Normally omit value_type for inference. Explicit types are0=null,1=bool,2=int,3=float,4=string and must match. Integers use signed 32-bit range; other finite numeric values use floating point.

source_id defaults to user. create_only:true preserves an existing variable, so it does not establish that the new requested value was written. Readonly/system-source variables can reject changes. Temperature binding requires suitable numeric temperature variables; strings, invalid values, and stale data are not normal sensor readings.

Action templates

automation.actions.add/update saves reusable actions. This example logs a message rather than controlling hardware.

JSON Source: example-021.json

```
{
  "id": "example-action",
  "active_version": "CURRENT_VERSION",
  "name": "Record a message",
  "enabled": true,
  "type": "log",
  "async": false,
  "delay_ms": 0,
  "log": {
    "level": 3,
    "message": "Example action"
  }
}
```

Shared fields: id up to 63 bytes, name 47 bytes, description 63 bytes; enabled defaults true, async false; delay_ms0-65535 milliseconds. created_at is an optional nonnegative integer time value, normally retained from the existing object.

type	Type-specific subobject and fields
log	log: level, message up to 47 bytes
set_var	set_var: variable, value, optional value_type
ssh_cmd	ssh: host_ref, command, timeout_ms, async

type	Type-specific subobject and fields
ssh_cmd_ref	ssh_ref: cmd_id referencing a saved command
gpio	gpio: pin, level as a **boolean**, pulse_ms
device_ctrl	device: device, action using supported actions
cli	cli: command, var_name, timeout_ms
webhook	webhook: url, method, body_template
led	led: device, ctrl_type and fields for that display function

Host references, variable names, command references, and device/action names are generally up to 63 bytes. Inline SSH/CLI commands and webhook URL are up to 95 bytes, method 7 bytes, body_template 63 bytes. Overlong fields are rejected; do not rely on truncation. Inline action commands have less capacity than saved ssh.commands.

LED action fields

led.ctrl_type is fill/effect/brightness/off/text/image/qrcode/filter/filter_stop/text_stop. Supply fields for the chosen function, not every parameter from a direct LED API.

Shared display fields: device, index (255=all), color string or r/g/b, brightness0-255, effect, speed, duration_ms0-65535. Matrix fields: text up to 127 bytes, font31, image_path127, qr_text255, qr_ecc string L/M/Q/H, filter31, filter_params, center, loop, scroll, align, x/y, auto_pos for text positioning. scroll is none/left/right/up/down; align left/center/right; x/y signed 16-bit coordinates.

QR foreground uses qr_fg; background uses qr_bg_image. Explicit black is an intentional color, not an omitted value. filter_params uses named numeric options supported by the effect. Execution additionally requires supported hardware and valid paths/configuration. Save/read before deliberately testing an effect.

Automation rules

automation.rules.add/update saves conditions and actions. Enabled rules can begin executing automatically; initially set enabled:false.

JSON Source: example-022.json

```
{
  "id": "example-rule",
  "active_version": "CURRENT_VERSION",
  "name": "Temperature example",
  "enabled": false,
  "logic": "and",
  "cooldown_ms": 10000,
  "conditions": [
    {
      "variable": "example.temperature",
      "operator": "gt",
      "value": 50
    }
  ],
  "actions": [
    {
      "type": "log",
      "level": 3,
      "message": "Temperature high"
    }
  ]
}
```

Field	Type and meaning
id / name / icon	Strings, at most63 /47 /63 bytes; ID rejects separators/control characters
enabled	Boolean, default true; example explicitly disables it
manual_trigger / show_on_dashboard / allow_manual_trigger	Booleans for triggering/presentation, not caller privilege
cooldown_ms	Unsigned 32-bit milliseconds
revision	Rule revision; use read revision as expected_revision for update conditions
logic / conditions	and/or, default and; up to 255 conditions
actions	Up to 255 entries; enabled rule cannot have no actions

A condition is {variable, operator, value}; operators eq/ne/lt/le/gt/ge/contains. Values follow the variable model. conditions:{logic, items:[...]} is also accepted; choose one representation.

Inline rule actions use **flat fields**: log has level/message, SSH has host_ref/command, GPIO has pin/level. Templates use nested subobjects instead. Inline SSH type is ssh; template type is ssh_cmd. A template_id reference takes precedence over cached inline fields.

Shared action fields: delay_ms, async, repeat_mode (once/while_true/count), repeat_count0-255, repeat_interval_ms0-65535, optional condition (condition model; null clears it). Select repetition with the action's effects in mind.

Inline type	Flat fields
log / set_var	level/message; or variable/value/value_type
ssh / ssh_cmd_ref	host_ref/command/ssh_async/timeout_ms; or cmd_id
gpio / device_ctrl	pin/boolean level/pulse_ms; or device/action
webhook / cli	url/method/body_template; or command/var_name/timeout_ms
led	Flat device/ctrl_type/r/g/b etc.; ctrl_type uses an integer enum rather than template strings

Prefer referencing a saved template for complex LED rules. Inline ctrl_type:0 fill,1 effect,2 brightness,3 off,4 text,5 image,6 qrcode,7 filter,8 filter_stop,9 text_stop. Inline qr_ecc is a character-code integer: L76/M77/Q81/H72.

filter_params remains its own object. Integer ranges: speed1-100; intensity/density/decay/amplitude0-255; scale1-100; levels 2-16; amount/saturation/frequency0-100; angle0-360; width1-16; wavelength1-32. These template/UI units differ from some direct LED filter units. Do not place an entire API request into the actions array.

Module configuration

config.module.show returns the module's config object. Read existing fields before config.module.set changes one key. Modules: net/dhcp/wifi/nat/led/fan/device/system. See individual operations for all support. key is local to the module; preserve the value type.

config.module.save and dirty state are separate from automation active_version. Networking changes can interrupt access; record the access route first. WiFi views below ADMIN omit passwords.

Monitoring widgets

ui.widgets stores frontend layout JSON; it does not create collectors. widgets is a frontend-interpreted array and may initially be empty. Start from ui.widgets.get and retain existing fields/IDs. refresh_interval is milliseconds, default 5000. Whole serialized configuration is limited to 4000 bytes. nvs_saved does not imply that a variable has data.

Import, deletion, and source selection

Use preview then apply with the preview's preconditions. tscfg is complete package JSON as a **string**, not an arbitrary object. Hosts, commands, sources, and actions import use expected_digest/expected_credential. Rule import has different confirmation fields; see its reference.

Preview deletion impact before selecting ordinary deletion, replacement, or cascade. Deleting definitions does not automatically stop remote processes or revoke remote keys. SD/internal-backup selection replaces the configuration set; use automation.config.preview/apply rather than overwriting protected files.

WebSocket integration

Connect to `ws://10.10.99.97/ws`, replacing the address with your device IP. This channel provides telemetry, logs, controller commands, and interactive SSH. A successful connection alone does not authorize private functions.

Authenticate and subscribe

Obtain token through HTTP login, then send:

JSON Source: example-023.json

```
{"type":"auth","token":"YOUR_TOKEN"}
```

Wait for auth_ok; failure returns auth_error. Ordinary telemetry topics are currently public, but establishing authentication remains useful for protected functions. Authentication cannot elevate a READ session to ROOT.

JSON Source: example-024.json

```
{"type":"subscribe","topic":"fan.status","params":{"interval":5000}}
```

Successful acknowledgement:

JSON Source: example-025.json

```
{"type":"subscribed","topic":"fan.status","success":true}
```

Illustrative data envelope:

JSON Source: example-026.json

```
{"type":"data","topic":"fan.status","timestamp":123,"data":{"fans":[]}}
```

timestamp is **seconds since start**; HTTP measurements and logs may use milliseconds. Subscription success is not proof that the first reading has arrived. If updates stop, inspect channel/source state and use HTTP readback as needed.

Topics

Topic	Default/minimum display interval	Content
system.dashboard	1000 ms	cpu/memory/info/network/power/fan/services; unavailable sections may be omitted
system.info	5000 ms	Controller information
system.memory	5000 ms	Controller memory
system.cpu	1000 ms	Controller CPU
network.status	5000 ms	Network status
power.status	5000 ms	Power status
fan.status	5000 ms	Fans
service.list	5000 ms	Controller services
device.status	Event-driven, default 2000 ms	Computer state events, not a periodic complete snapshot
ota.progress	Event-driven, default 1000 ms	Device-wide progress; check HTTP progress and the version after restart
config.pack.validated	Event-driven, default 0	Uploaded package validation, not configuration application

params.interval is a nonnegative integer in milliseconds. The first eight periodic topics reject0 and raise smaller intervals to their minimum. An interval does not convert the last three event topics into polling. Re-subscribing to the same topic on a connection updates that subscription; it does not create a history queue.

Unsubscribe with `{"type":"unsubscribe","topic":"fan.status"}` and inspect unsubscribed/success. `{"type":"ping"}` receives pong. Pong does not establish session authorization or healthy telemetry.

Logs

ADMIN or higher is required. Authenticate, then send log_subscribe. History also requires ADMIN.

Outgoing type	Fields	Result/effect
log_subscribe	minLevel, default VERBOSE	log_subscribed/minLevel, then type=log
log_set_level	minLevel	Changes this connection's filter, not all module output levels
log_get_history	limit default 500; minLevel ERROR; maxLevel VERBOSE	log_history with logs/total, bounded by buffer capacity

Outgoing type	Fields	Result/effect
log_unsubscribe	None	Stops this connection's log stream

Log entries contain timestamp in milliseconds since start, level, levelName, tag, message, and task. Levels 0-5 are NONE/ERROR/WARN/INFO/DEBUG/VERBOSE. Retention is bounded; reconnect does not automatically replay history.

Controller command terminal

This terminal controls the management controller, not the Linux system on either computer. ROOT is required and commands can change hardware/configuration.

Outgoing type	Fields	Result
terminal_start	token	Binds and requests the terminal; connected includes prompt; occupancy returns terminal_busy/retry_after
terminal_input	data: complete command string	output/data messages, then done with code/output_incomplete/stopped
terminal_interrupt	None	Requests interruption; command_stopping may precede done
terminal_stop	None	Releases this connection's terminal, not a rollback of external effects

Only one terminal owner is active at a time. Do not take over a live terminal. Missing output does not by itself prove execution failure. Prefer structured HTTP state APIs over parsing terminal output for a first integration.

Interactive SSH Shell

ROOT is required. ssh_connect carries token and binds the connection. Verify remote fingerprints first; this Shell has no automatic unknown-fingerprint acceptance switch.

Outgoing type	Fields	Result
ssh_connect	token, host, user; optional port22/password for this remote login	ssh_status connecting/connected/error; this entry uses password authentication
ssh_input	data	Writes text; explicitly include required newlines
ssh_resize	width, height	Integers1-65535 for terminal columns/rows
ssh_signal	signal	Sends a supported signal; unsupported requests fail
ssh_disconnect	None	Disconnects this connection's Shell; observe final ssh_status

Output uses ssh_output/data, status uses ssh_status/status/message. A connection controls only its own Shell. Do not replay input automatically after a failed control or disconnect; the remote side may have received it.

HTTP-started streaming SSH

ssh.exec_stream starts through HTTP and returns session_id/boot_id/result_key. It is separate from the interactive Shell. Authenticate WebSocket before starting to avoid missing early events. Private events are delivered only to connections authorized for the initiating session.

Incoming type	Fields and interpretation
ssh_exec_start	session_id, command and patterns; started, not completed
ssh_exec_output	session_id, output_offset, output_length, is_stderr, data_b64
ssh_exec_match	session_id, expect_matched/fail_matched, optional extracted/is_final
ssh_exec_running	In-progress response from HTTP status query
ssh_exec_done	session_id, execution_status, exit_code, status/success and matches
ssh_exec_cancelled	session_id, execution_status; cancellation does not establish remote process termination
ssh_exec_error	session_id, error, optional stage/detail

Decode data_b64 into bytes, verify output_length, and use output_offset to detect gaps/duplicates. Stream UTF-8 decoding across frames; characters can span messages.

Terminal results may include output_bytes/output_incomplete/collection_truncated/conditions_complete. **Execution completion, condition matching, and output completeness are separate.** exit_code=0 can coexist with missing output; incomplete matching is not a positive result. After disconnect, query ssh.exec_status with the receipt; it does not rerun the command or promise all historical output.

Capacity and reconnection

This build allows eight WebSocket clients and 32 topic subscriptions service-wide. Large frames, slow clients, exclusive tasks, and shutdown can reject work. Input must be protocol JSON text, at most 64 KiB; do not use fragmentation as a file-upload mechanism. JSON depth limits also apply.

Reconnect with delay, reauthenticate, and restore subscriptions. Do not automatically repeat terminal_input, ssh_input, or HTTP controls. Display last-update age and stale/disconnected state so old values do not appear current.

The companion subscribe.js example uses a browser that can reach the device and needs no third-party libraries.

File transfer and updates

Ordinary files and APP/WWW images use dedicated binary routes. Bodies contain original file bytes; successful responses do not necessarily include code. These routes are separate from the 277 logical API names.

Ordinary files

Route	Method/access	Request and result
/api/v1/file/download	GET, READ; ADMIN for sensitive paths	Absolute path in query; response is file bytes, not a directory
/api/v1/file/upload	POST, ADMIN	Destination path in query, original bytes in body; success returns path/size/status=uploaded

HTTP uploads only allow validated ordinary files under `/sdcard/`. Other file operations, such as storage.rename/delete, have a different path scope; they do not allow HTTP upload to `/spiffs/` or `/www/`. Sensitive configuration and private-key reads require ADMIN. Managed automation directories, aliases, backups, and temporary files remain protected; edit them through object configuration APIs.

Uploads allow at most 16 MiB with a 300-second receive budget, also limited by memory, free space, and storage state. Supply one complete nonempty file. Repeated requests do not append chunks. Upload replaces an existing destination; preserve needed content first. URL-encode paths once.

Configure TS_BASE and TS_TOKEN before running:

SH Source: example-027.sh

```
curl --connect-timeout 5 --max-time 330 \
  -H "Authorization: Bearer $TS_TOKEN" \
  -H 'Content-Type: application/octet-stream' \
  --data-binary @example.txt \
  "$TS_BASE/api/v1/file/upload?path=/sdcard/example.txt"

curl --connect-timeout 5 --max-time 60 \
  -H "Authorization: Bearer $TS_TOKEN" \
  --get --data-urlencode 'path=/sdcard/example.txt' \
  "$TS_BASE/api/v1/file/download" -o downloaded-example.txt
```

A .tscfg upload adds config_pack.valid/reason/signature information. uploaded means saved, not trusted or applied. Preview and apply through the object-specific import; config.pack.apply does not apply generic configuration. Ordinary data packages and command-bearing rule packages have different signing-usage checks; ordinary validation does not replace rule-import validation.

For file_replace_uncertain or a lost connection, inspect file metadata and download its contents before retrying.

Prepare images

Obtain a complete APP firmware image and matching www.bin for the hardware, partition layout, and firmware version. A WWW image is not an individual HTML file. Back up configuration, ensure stable power and a maintenance window, and prepare a recovery path.

Downloads accept HTTP and HTTPS. HTTPS uses system certificate trust by default. skip_verify is a download option; it cannot fix damaged or incompatible images or insufficient capacity. Normally keep it false. Current start APIs do not accept a private-CA cert_pem parameter.

Choose a transfer method

Method	API	Requirements
Device downloads	ota.start_url / ota.www.start	Device can reach the image server; check network and storage
SD file	ota.start_file / ota.www.start_sdcard	Mounted SD with complete image files
Client uploads	Dedicated binary routes below	Client has complete files; device has writable SD for staging and recovery cache

Route	Method/access	Options and success
/api/v1/ota/firmware	POST, ADMIN	Complete APP body; auto_reboot query defaults true; status=success, size, message, reboot_pending, auto_reboot_requested
/api/v1/ota/www	POST, ADMIN	Complete www.bin body; status=success, size, message; no automatic restart

Both allow at most 16 MiB with a 300-second receive budget, subject to target-partition capacity and memory. Firmware headers and installation completeness are checked. Successful upload saves SD recovery files and a manifest with automatic recovery disabled; application acceptance after restart remains a separate step.

This example disables automatic restart so the maintenance process can inspect the result first:

SH Source: example-028.sh

```
curl --connect-timeout 5 --max-time 600 \
  -H "Authorization: Bearer $TS_TOKEN" \
  -H 'Content-Type: application/octet-stream' \
  --data-binary @firmware.bin \
  "$TS_BASE/api/v1/ota/firmware?auto_reboot=false"
```

firmware.bin is a client-side filename; replace it with the complete firmware image. This request writes to the device. With auto_reboot=true, restart is requested about 2 seconds after a successful response and the connection may close.

Confirm progress and completion

After URL or SD initiation, read ota.progress, or ota.www.progress for WWW. Progress is device-wide without a private per-login task receipt. Coordinate updates through one maintenance process to avoid concurrent starts or cancelling another update.

SH Source: example-029.sh

```
curl --connect-timeout 5 --max-time 15 \
  -H "Authorization: Bearer $TS_TOKEN" \
  "$TS_BASE/api/v1/ota/progress"
```

APP progress has state/percent/received_size/total_size/message. WWW uses received/total and adds running. pending_reboot means a restart is needed; error means failure. 100 percent alone does not prove the new version is running. To cancel, POST an empty object to ota.abort or ota.www.abort, then confirm state before starting again.

For binary uploads, use the HTTP response rather than assuming global progress identifies this request. HTTP409 with status=partial, error=recovery_cache_failed, and flash_written=true means Flash was written but the SD recovery cache was not fully saved. Inspect partitions and recovery files. A lost response also does not establish that no write occurred.

After firmware installation, restart as planned, log in again, inspect ota.status/partitions, and verify application health. ota.validate confirms the current firmware; call it after health checks.

Rollback and SD recovery

Check ota.partitions.can_rollback before ota.rollback. Rollback may restart immediately. Reconnect and verify version, configuration compatibility, and application state.

SD recovery uses TianShanOS.bin, www.bin, and manifest.json under `/sdcard/recovery/`. armed controls whether recovery is enabled; force controls version comparison. Existing files do not establish enabled recovery. Follow the publisher's recovery procedure because recovery affects a later boot and persistent storage; uploading files alone does not prove success.

No recovery preparation, status, or authorization API is exposed. Do not invent routes or treat ota.upload_begin/end/abort as a complete segmented-transfer protocol. Ordinary integrations should use the binary routes above.

Keep an integration running

Manage login state

Keep one session per client process and reuse it for requests. Schedule renewal from expires_in; log in again after a clear authentication failure rather than on every poll. There are eight session slots shared by scripts and browsers. When slots are full, a new login may invalidate an older session for the same account.

Normal sessions expire after 86400 seconds in this build. Restart, logout, password changes, and revocation require reauthentication. Treat tokens as opaque credentials, not values to parse or construct.

Poll or subscribe

Use HTTP for occasional reads and WebSocket for a continuously displayed dashboard. Subscribe only to required topics and unsubscribe when leaving a view. Display update intervals do not change hardware sampling or fan control periods.

After a disconnect, reconnect, authenticate, and restore subscriptions. Do not automatically repeat power commands, SSH execution, or updates. Telemetry and logs missed during disconnection are not guaranteed to be replayed.

Track one operation

Streaming SSH uses session_id+boot_id+result_key; the execution session_id differs from the login session_id. OTA uses device-wide progress without a private operation receipt.

SSH result queries require the complete receipt and initiating login; cancellation may use just the execution session_id from that same login. Do not share receipts. Restart, replacement, or limited retention can make results unavailable; inspect the remote state then. Coordinate OTA start, cancellation, and progress through one maintenance process, since global state is not an exclusive private task.

Timeouts and retries

Set connection and execution deadlines separately. Scans, network diagnostics, remote commands, and updates may take longer than ordinary reads. An HTTP timeout does not cancel server-side work.

Retry queries with bounded concurrency and increasing delay. For an uncertain write, inspect state first; repeat only when you know it did not start and still need it. Prefer an explicit desired state over toggle.

Configuration changes

Read active_version before writing and refresh after a conflict. Rules may also require expected_revision. Do not overwrite complete objects from a filtered list response.

Record application version, build date/time, and the capabilities used by your integration. Identical version strings can identify different repaired builds. After an upgrade, review method, role, field, and deprecation changes.

Operational records

Record time, target, operation type, business code, a brief reason, and non-secret operation identifiers. Store result credentials in controlled storage if retention is necessary. Ordinary logs should not contain Authorization, login bodies, private keys, or package secrets.

Troubleshooting

Record method, path, HTTP status, business code, reason, and operation receipt. Keep passwords, tokens, private keys, and full authorization headers out of support tickets.

Cannot connect

Use the current HTTP device address, such as http://10.10.99.97. The default port is80 and need not be written explicitly. Check the address, network reachability, and WebUI service state.

A working mTLS diagnostic page on443 does not prove the business API is running. Once the public system.info request works, investigate login and the target feature.

Authentication or permission failure

HTTP401 usually means a missing, invalid, or expired session;403 usually means insufficient privilege. Some handlers report authorization errors as business errors within HTTP200.

Log in again and inspect data.level. Admin can use ADMIN configuration and maintenance APIs; raw SSH execution, automation definition writes, and selected account management still require ROOT. Initial default passwords are rm01; use the current password if it has been changed. Password changes revoke affected sessions. Wait out login lockout instead of launching repeated concurrent retries.

Missing temperatures or telemetry

Inspect temp.sources for validity, update time, age, and source. Variable temperature input can be partly stale, so check valid weights/counts. monitor.data depends on a monitoring service on the compute host; inspect monitor connection and address configuration. Powered-on hardware does not guarantee telemetry has arrived.

Accepted but unfinished

Preserve and query the receipt or run record. SSH requires final status, exit code, and output completeness; managed services require verified state; OTA requires device-wide progress, partition state, and the running version after restart. started, cancel_requested, or a historical record alone is insufficient.

Configuration changed

For configuration_changed or a version conflict, reread current state and the relevant object. Compare and merge your unsaved input before submitting the new version. Do not simply replace the version token and repeat a delete/import/overwrite request.

Save failed or outcome unknown

Inspect applied, persisted, saved, uncertain, result_unknown, and the reason. Runtime state can change even when persistence fails. Read it back, then check saved startup configuration or backups as applicable. A network failure does not establish that the request had no effect.

SSH fingerprint changed

Stop automatic retries and obtain the current fingerprint from the server administrator. Check for reinstall, replacement, or address reassignment. Use the documented confirmation fields only after verification. An identical address does not guarantee the same host.

Busy or at capacity

Reduce concurrency, reuse sessions, and close unused subscriptions or terminals. Inspect active work. Tasks cannot queue indefinitely because a client keeps retrying. Wait for your own task's stop sequence; do not take over another connection's work.

Upload or update failed

Check role, complete request delivery, path, SD status, free capacity, target partition size, and complete images for the hardware/partition layout. Uploaded does not mean package validation or installation succeeded. For SSH, query the retained receipt; for OTA, inspect global progress and partition state before retrying, rolling back, or deleting recovery material.

Version and compatibility

This guide is based on **TianshanOS 0.6.4**. It remains useful for later versions where interfaces and procedures are unchanged. If parameters or results differ, follow the documentation for your installed version.

Connections and accounts

Item	Usage
Business API	HTTP, default port 80, prefix /api/v1; e.g. http://10.10.99.97
Live messages	WebSocket, ws://device-address/ws
Default accounts	admin/root, both initially rm01; use the current password if changed
Session authentication	Login token in Authorization: Bearer
Access control	Minimum role per endpoint, with additional resource/version/ownership checks

Feature prerequisites

An available endpoint does not establish that all runtime dependencies are ready. Inspect the relevant status before use.

Feature	Prerequisites
Temperature, fans, power, computers	Initialized drivers, existing targets, valid measurements
Compute telemetry	Running monitor, reachable host service, fresh valid data
Files and SD updates	Required storage mounted, file present, sufficient capacity; direct upload has a separate flow
SSH/SFTP and managed services	Network, credentials, host fingerprints, keys, saved configuration, task capacity
Automation	Loaded/enabled configuration, correct engine state, valid references, matching write versions
Configuration packages	Device/recipient identity and publisher trust; export also requires Developer capability
Device PKI and updates	Valid PKI material/time for diagnostics; complete compatible APP/WWW images for updates
Matrix content	Supported target and usable font/image files

Aliases and unsupported entries

config.pack.apply is retained but does not apply generic configuration; use the object-specific import. led.set writes pixel ranges. ota.upload_begin/end/abort control an upload, but no public upload_write data API is provided; ordinary integrations should use complete-image binary upload.

OTA retains upgrade_url/start_url and upgrade_file/start_file aliases. Prefer start_url/start_file for new integrations.

mTLS diagnostic service

This separate certificate-authenticated service defaults to 443. It is not another port for /api/v1. Availability depends on PKI material, configuration, and service state. Bearer tokens do not replace client certificates.

Method/path	Minimum certificate role	Purpose
GET /	anonymous	Health
GET /api/auth/whoami	viewer	Certificate identity
GET /api/test/viewer	viewer	Read-role test
GET /api/test/operator	operator	Operator-role test
GET /api/test/admin	admin	Admin read test

Method/path	Minimum certificate role	Purpose
POST /api/test/admin-action	admin	Admin write test

These endpoints diagnose health, identity, and permissions rather than performing device business actions. Certificate viewer/operator/admin and business READ/ADMIN/ROOT belong to separate identity systems and cannot be interchanged.

Included examples

The PDF includes read_status.py and subscribe.js as attachments. Save them using a PDF reader that supports attachments; the editable companion contains the same files. Placeholder addresses, tokens, and certificate paths are not actual device settings.

Copy a code example

Code boxes contain selectable text without line numbers mixed into the code. To preserve indentation and line endings exactly, use the paperclip in the upper-right corner to save the named source file, or save it from your PDF reader's attachment panel.

If your reader does not support attachments, the download package contains all original examples in code-examples/en with matching filenames. Save API request JSON as request.json for the corresponding endpoint. Response examples and WebSocket messages serve the purposes described in their chapters. Fill in addresses, tokens, and other placeholders before use.

Python read-only query

Python 3 and its standard library are sufficient. Use your device address and run:

SH Source: example-030.sh

```
python3 read_status.py --base 'http://10.10.99.97' \
  --user admin
```

Enter the password at the hidden prompt rather than in command history. The script logs in, reads status, and attempts logout. It does not control hardware, run remote tasks, or upgrade. It follows no redirects and performs no automatic retries. Check valid/duty_valid and other measurement flags in its output.

Browser subscription

subscribe.js exports watchTianShan as an ES module without third-party libraries. Use a browser that can reach the device. Obtain a token through HTTP login first:

JAVASCRIPT Source: example-031.js

```
import {watchTianShan} from './subscribe.js';
const close = watchTianShan({
  baseUrl: 'http://10.10.99.97',
  token: yourLoginToken,
  onData: message => renderYourDashboard(message.data),
  onState: state => showConnectionState(state)
});
```

Your page supplies renderYourDashboard/showConnectionState. Call close() when leaving. Mark readings stale after disconnect before restoring authentication/subscriptions. Keep tokens out of source, URLs, and ordinary logs. The example does not reconnect automatically or replay actions.

Complete API index

All 277 registered names are included. Select a name/page to jump; each entry lists methods,access,parameters,and completion checks.

Send reference examples as POST JSON. Save as request.json,replace placeholders,and use the listed path. Public calls may omit Authorization.

SH Source: example-032.sh

```
curl --connect-timeout 5 --max-time 60 \
  -H "Authorization: Bearer $TS_TOKEN" \
  -H 'Content-Type: application/json' --data-binary @request.json \
  "$TS_BASE/api/v1/PATH_FROM_REFERENCE"
```

Writes have real effects. Obtain required versions,IDs,certificates,or release materials first. Use GET only where allowed; prefer POST JSON for complex values.

Alphabetical locator

Labels indicate minimum entry access; resource,session,and feature conditions still apply. Retained unsupported entries are marked in the reference.

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Accounts and sessions APIs

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Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
auth.admin.reset_password	Restore the admin default password
auth.admin.set_password	Root sets admin password
auth.change_password	Change password
auth.login	User login
auth.logout	User logout
auth.root.set_password	Root sets root password
auth.status	Check auth status

auth.admin.reset_password

Restore the admin default password

Methods: POST
Path: `/api/v1/auth/admin/reset_password`
Minimum access: ROOT

Parameters

Parameter	Meaning
`token` string · Required	Valid login token in JSON; session_id is not a substitute. If omitted: No default

Request example

JSON Source: example-033.json

```
{"token":"ROOT_TOKEN"}
```

Response and completion

data.success/username/password_changed. After success, admin logs in with default password rm01.

Usage notes

Root restores the admin password to rm01. Use that default password after reset.

auth.admin.set_password

Root sets admin password

Methods: POST
Path: `/api/v1/auth/admin/set_password`
Minimum access: ROOT

Parameters

Parameter	Meaning
`token` string · Required	Valid login token in JSON; session_id is not a substitute. If omitted: No default
`new_password` string · Required	New password, 4-64 bytes. If omitted: No default

Request example

JSON Source: example-034.json

```
{"token":"ROOT_TOKEN","new_password":"NEW_PASSWORD"}
```

Response and completion

data.success/sessions_revoked/reauth_required/username/password_changed. Affected-account sessions are revoked.

Usage notes

Both the Authorization identity and the JSON token must pass root checks; both fields have a purpose.

auth.change_password

Change password

Methods: POST
Path: `/api/v1/auth/change_password`
Minimum access: Body token (authentication-specific)

Parameters

Parameter	Meaning
`token` string · Required	Valid login token in JSON; session_id is not a substitute. If omitted: No default
`old_password` string · Required	Current password. If omitted: No default
`new_password` string · Required	New password, 4-64 bytes. If omitted: No default

Request example

JSON Source: example-035.json

```
{"token":"YOUR_TOKEN","old_password":"CURRENT_PASSWORD","new_password":"NEW_PASSWORD"}
```

Response and completion

data.success/sessions_revoked/reauth_required. Log in again after success.

Usage notes

The account holder chooses when to change the password. The old session is invalidated after success.

auth.login

User login

Methods: POST
Path: `/api/v1/auth/login`
Minimum access: Public

Parameters

Parameter	Meaning
`username` string · Required	Configured admin or root account. If omitted: No default
`password` string · Required	Current password for that account. If omitted: No default

Request example

JSON Source: example-036.json

```
{"username":"admin","password":"rm01"}
```

Response and completion

POST success has code=0 and data.token/session_id/username/level/expires_in(seconds)/password_changed.

Usage notes

Five consecutive failures lock login for 300 seconds. Login does not establish WebSocket subscriptions.

auth.logout

User logout

Methods: POST
Path: `/api/v1/auth/logout`
Minimum access: Body token (authentication-specific)

Parameters

No business parameters.

Request example

JSON Source: example-037.json

```
{}
```

Response and completion

Dedicated POST returns {success:true}. Also discard the client token.

Usage notes

Authorization identifies the session to close; no JSON is needed. The internal handler's body-token form is not the dedicated HTTP POST contract.

auth.root.set_password

Root sets root password

Methods: POST
Path: `/api/v1/auth/root/set_password`
Minimum access: ROOT

Parameters

Parameter	Meaning
`token` string · Required	Valid login token in JSON; session_id is not a substitute. If omitted: No default
`new_password` string · Required	New password, 4-64 bytes. If omitted: No default

Request example

JSON Source: example-038.json

```
{"token":"ROOT_TOKEN","new_password":"NEW_PASSWORD"}
```

Response and completion

data.success/sessions_revoked/reauth_required/username/password_changed. Affected-account sessions are revoked.

Usage notes

Both the Authorization identity and the JSON token must pass root checks; both fields have a purpose.

auth.status

Check auth status

Methods: POST
Path: `/api/v1/auth/status`
Minimum access: Body token (authentication-specific)

Parameters

Parameter	Meaning
`token` string · Required	Valid login token in JSON; session_id is not a substitute. If omitted: No default

Request example

JSON Source: example-039.json

```
{"token":"YOUR_TOKEN"}
```

Response and completion

data.valid reports validity; valid sessions include username/level/expires_in/password_changed. valid=false can coexist with code=0.

Controller status APIs

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Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
system.apis	List all registered APIs
system.cpu	Get CPU core statistics
system.info	Get system information
system.log.level	Get/set log level
system.memory	Get memory information
system.memory_detail	Get detailed memory analysis
system.reboot	Reboot the system
system.tasks	Get task list

system.apis

List all registered APIs

Methods: GET / POST
Path: `/api/v1/system/apis`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-040.json

```
{}
```

Response and completion

data.apis is an array of name strings with total; legacy monitor-presence flags may also appear.

Usage notes

A name list does not replace parameter/access documentation.

system.cpu

Get CPU core statistics

Methods: GET / POST
Path: `/api/v1/system/cpu`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-041.json

```
{}
```

Response and completion

data.total_usage/cores; each core includes id/usage. These are controller metrics, not inference-computer metrics.

system.info

Get system information

Methods: GET / POST
Path: `/api/v1/system/info`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-042.json

```
{}
```

Response and completion

data.app has name/version/idf_version/compile_time/compile_date; chip has model/cores/revision/features; flash_size is bytes and uptime_ms milliseconds.

system.log.level

Get/set log level

Methods: POST / PUT / DELETE
Path: `/api/v1/system/log/level`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`level` integer · Optional	0=NONE,1=ERROR,2=WARN,3=INFO,4=DEBUG,5=VERBOSE. If omitted: Unchanged
`tag` string · Optional	Change only this module's log level; omission affects all modules. If omitted: Global

Request example

JSON Source: example-043.json

```
{"level":3}
```

Response and completion

data.status; a change also includes level/tag.

Usage notes

Intended for temporary diagnostics; persistent file logging is not promised.

system.memory

Get memory information

Methods: GET / POST
Path: `/api/v1/system/memory`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-044.json

```
{}
```

Response and completion

data.free_heap/min_free_heap are bytes. internal and available psram contain total/free/allocated/largest_block, all in bytes.

system.memory_detail

Get detailed memory analysis

Methods: GET / POST
Path: `/api/v1/system/memory_detail`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-045.json

```
{}
```

Response and completion

data.dram/psram/iram/dma/static/rtc/nvs/tasks/history/tips diagnostic groups. total/free/used are memory statistics; stack_allocation_estimated identifies estimation.

Usage notes

Estimates in one diagnostic snapshot are not an exact physical-memory breakdown or proof of a leak.

system.reboot

Reboot the system

Methods: POST / PUT / DELETE
Path: `/api/v1/system/reboot`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`delay` integer · Optional	Delay in milliseconds, ultimately bounded to 100-10000. If omitted: 500

Request example

JSON Source: example-046.json

```
{"delay":500}
```

Response and completion

data.status=rebooting, delay_ms, and waiting_for_maintenance=false. Restart is requested after the delay; this API does not wait for an update to finish.

Usage notes

Interrupts management connectivity. Save required settings, wait for recovery, and log in again.

system.tasks

Get task list

Methods: GET / POST
Path: `/api/v1/system/tasks`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-047.json

```
{}
```

Response and completion

data.tasks/count; tasks include name/state/priority/core/stack_hwm. stack_hwm is a native stack watermark, not total free memory.

Fans APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
fan.config	Get fan configuration including curve
fan.curve	Set temperature curve for fan
fan.enable	Enable or disable a fan
fan.limits	Set duty cycle limits (min/max)
fan.mode	Set fan operating mode
fan.save	Save fan configuration to NVS
fan.set	Set fan speed (manual mode)
fan.status	Get fan status

fan.config

Get fan configuration including curve

Methods: GET / POST
Path: `/api/v1/fan/config`
Minimum access: READ

Parameters

Parameter	Meaning
`id` integer · Optional	Fan ID; obtain it from fan.status first. If omitted: All when omitted

Request example

JSON Source: example-048.json

```
{"id":0}
```

Response and completion

One/all configurations: id/mode/min_duty/max_duty/invert_pwm/hysteresis/min_interval/curve. Curve points use temp(Celsius)/duty(percent).

fan.curve

Set temperature curve for fan

Methods: POST / PUT / DELETE
Path: `/api/v1/fan/curve`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` integer · Required	Fan ID within the device's supported range. If omitted: No default
`curve` array · Required	At most 10 {temp:Celsius,duty:percent} points; ascending temperatures and duty 0-100. If omitted: No default
`hysteresis` number · Optional	Temperature hysteresis, 0-20 Celsius. If omitted: Current value
`min_interval` integer · Optional	Minimum adjustment interval, integer 100-60000 milliseconds. If omitted: Current value

Request example

JSON Source: example-049.json

```
{"id":0,"curve":[{"temp":30,"duty":30},{"temp":60,"duty":80}],"hysteresis":3,"min_interval":2000}
```

Response and completion

data.id/points/applied/persisted.

Usage notes

Curve configuration and control mode are separate. Read back configuration and select curve mode if needed.

fan.enable

Enable or disable a fan

Methods: POST / PUT / DELETE
Path: `/api/v1/fan/enable`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` integer · Required	Fan ID within the device's supported range. If omitted: No default
`enable` boolean · Required	true enables; false disables. If omitted: No default

Request example

JSON Source: example-050.json

```
{"id":0,"enable":true}
```

Response and completion

data.id/enabled.

Usage notes

Disabling can reduce cooling. Check temperature and protection separately.

fan.limits

Set duty cycle limits (min/max)

Methods: POST / PUT / DELETE
Path: `/api/v1/fan/limits`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` integer · Required	Fan ID within the device's supported range. If omitted: No default
`min_duty` integer · Optional	Minimum duty, 0-100 percent, no greater than maximum. If omitted: Current value
`max_duty` integer · Optional	Maximum duty, 0-100 percent. If omitted: Current value

Request example

JSON Source: example-051.json

```
{"id":0,"min_duty":20,"max_duty":100}
```

Response and completion

data.id/min_duty/max_duty/applied/persisted.

fan.mode

Set fan operating mode

Methods: POST / PUT / DELETE
Path: `/api/v1/fan/mode`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` integer · Required	Fan ID within the device's supported range. If omitted: No default
`mode` string · Required	off/manual/auto/curve. If omitted: No default

Request example

JSON Source: example-052.json

```
{"id":0,"mode":"auto"}
```

Response and completion

data.id/mode/applied/persisted. Distinguish runtime application from storage failure.

fan.save

Save fan configuration to NVS

Methods: POST / PUT / DELETE
Path: `/api/v1/fan/save`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` integer · Optional	Fan ID. If omitted: All fans

Request example

JSON Source: example-053.json

```
{}
```

Response and completion

data.status/persisted. Inspect failure details; do not assume retention after restart.

fan.set

Set fan speed (manual mode)

Methods: POST / PUT / DELETE
Path: `/api/v1/fan/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` integer · Required	Fan ID within the device's supported range. If omitted: No default
`duty` integer · Required	Requested duty, 0-100 percent. If omitted: No default

See the shared data model.

Request example

JSON Source: example-054.json

```
{"id":0,"duty":50}
```

Response and completion

Applied fan state; inspect duty_valid/duty/target_duty/mode.

Usage notes

Affects cooling. Change deliberately and read back actual/requested output.

fan.status

Get fan status

Methods: GET / POST
Path: `/api/v1/fan/status`
Minimum access: Public

Parameters

Parameter	Meaning
`id` integer · Optional	Fan ID; obtain it from fan.status first. If omitted: All when omitted

See the shared data model.

Request example

JSON Source: example-055.json

```
{"id":0}
```

Response and completion

With id, data is one fan; otherwise data.fans plus temperature/temp_valid/temp_source. See the fan response model.

Temperature APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
temp.bind	Bind/unbind temperature to a system variable
temp.manual	Set/get manual temperature mode
temp.read	Read current temperature
temp.select	Set/get preferred temperature source
temp.sources	Get all temperature sources info
temp.status	Get temperature system status

temp.bind

Bind/unbind temperature to a system variable

Methods: POST / PUT / DELETE
Path: `/api/v1/temp/bind`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`variables` array · Optional	At most 8 {name,weight} entries; an empty array clears multi-variable bindings. If omitted: Unchanged
`variable` string / null · Optional	Legacy single-variable binding; null or empty string clears it. If omitted: Unchanged

Request example

JSON Source: example-056.json

```
{"variables":[{"name":"compute.temperature","weight":1}]}
```

Response and completion

data.bound_variables, binding state, valid counts/weights, weighted_temp_c, partial_stale/valid. Unauthorized variable values are filtered.

Usage notes

Variables must exist and be numeric temperatures. ADMIN may query with binding fields omitted; supplying variables or variable to change or clear bindings requires ROOT.

temp.manual

Set/get manual temperature mode

Methods: POST / PUT / DELETE
Path: `/api/v1/temp/manual`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`enable` boolean · Optional	Enable or disable manual temperature mode. If omitted: Unchanged
`temperature_c` number · Optional	Celsius, -50.0 to 150.0. If omitted: Unchanged
`temperature` number · Optional	Alias of temperature_c; preferably send only one temperature field. If omitted: Unchanged
`temperature_01c` integer · Optional	Tenths of Celsius, -500 to 1500; used when no Celsius field is supplied. If omitted: Unchanged

Request example

JSON Source: example-057.json

```
{"enable":true,"temperature_c":45}
```

Response and completion

data.manual_mode/current_temp_c/current_temp_01c/active_source.

Usage notes

Changes temperature input used by control. Explicitly disable manual mode and read back state after testing.

temp.read

Read current temperature

Methods: GET / POST
Path: `/api/v1/temp/read`
Minimum access: READ

Parameters

Parameter	Meaning
`source` string · Optional	default/local or sensor_local/agx or agx_auto/variable/manual. If omitted: Current active source

See the shared data model.

Request example

JSON Source: example-058.json

```
{}
```

Response and completion

data.temperature_c/temperature_01c/source/timestamp_ms/valid; variable sources may include partial_stale and binding counts/weights.

temp.select

Set/get preferred temperature source

Methods: POST / PUT / DELETE
Path: `/api/v1/temp/select`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`source` string · Optional	auto/agx/sensor/variable. If omitted: Query only

Request example

JSON Source: example-059.json

```
{"source":"auto"}
```

Response and completion

data.preferred_source/active_source/temperature_c/valid.

temp.sources

Get all temperature sources info

Methods: GET / POST
Path: `/api/v1/temp/sources`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-060.json

```
{}
```

Response and completion

Current temperature/validity plus providers. Entries include name/type/active/valid/stale/last_value_c/last_value_01c/last_update_ms/age_ms/update_count.

Usage notes

age_ms=null means no usable update time. Variable sources also report partial failures and valid counts/weights.

temp.status

Get temperature system status

Methods: GET / POST
Path: `/api/v1/temp/status`
Minimum access: Public

Parameters

No business parameters.

See the shared data model.

Request example

JSON Source: example-061.json

```
{}
```

Response and completion

data.initialized/manual_mode/active_source/preferred_source/temperature_c/temperature_01c/timestamp_ms/valid, possibly bound_variable.

Power and protection APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
power.chip	Get power chip data
power.debug	Set debug mode
power.interval.set	Set sampling interval
power.monitor.start	Start power monitoring
power.monitor.stop	Stop power monitoring
power.protection.config	Get voltage protection configuration
power.protection.set	Configure voltage protection
power.protection.status	Get voltage protection status
power.stats	Get monitoring statistics
power.stats.reset	Reset monitoring statistics
power.status	Get power monitoring status
power.threshold.set	Set voltage thresholds
power.voltage	Read current voltage

power.chip

Get power chip data

Methods: GET / POST
Path: `/api/v1/power/chip`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-062.json

```
{}
```

Response and completion

data.voltage_v/current_a/power_w/valid/crc_valid/timestamp_ms/raw_data. Use only valid, integrity-checked values.

power.debug

Set debug mode

Methods: POST / PUT / DELETE
Path: `/api/v1/power/debug`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`enable` boolean · Required	Enable debug output. If omitted: No default

Request example

JSON Source: example-063.json

```
{"enable":false}
```

Response and completion

data.debug_enabled.

power.interval.set

Set sampling interval

Methods: POST / PUT / DELETE
Path: `/api/v1/power/interval/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`interval_ms` integer · Required	Sampling interval in milliseconds; the device rejects values below its supported range. If omitted: No default

Request example

JSON Source: example-064.json

```
{"interval_ms":1000}
```

Response and completion

data.interval_ms.

power.monitor.start

Start power monitoring

Methods: POST / PUT / DELETE
Path: `/api/v1/power/monitor/start`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-065.json

```
{}
```

Response and completion

data.running.

Usage notes

Stopping sampling affects future updates. Check the data required by protection as well.

power.monitor.stop

Stop power monitoring

Methods: POST / PUT / DELETE
Path: `/api/v1/power/monitor/stop`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-066.json

```
{}
```

Response and completion

data.running.

Usage notes

Stopping sampling affects future updates. Check the data required by protection as well.

power.protection.config

Get voltage protection configuration

Methods: GET / POST
Path: `/api/v1/power/protection/config`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-067.json

```
{}
```

Response and completion

data.low_voltage_threshold/recovery_voltage_threshold(V), shutdown_delay_sec/recovery_hold_sec/fan_stop_delay_sec(seconds), and defaults.

power.protection.set

Configure voltage protection

Methods: POST / PUT / DELETE
Path: `/api/v1/power/protection/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`enable` boolean · Optional	Start/stop protection; supplying this field always saves enablement. If omitted: Unchanged
`low_threshold` number · Optional	Low-voltage threshold in V. If omitted: Unchanged
`recovery_threshold` number · Optional	Recovery threshold in V. If omitted: Unchanged
`shutdown_delay` integer · Optional	Low-voltage shutdown delay in seconds. If omitted: Unchanged
`recovery_hold` integer · Optional	Required recovery hold time in seconds. If omitted: Unchanged
`fan_stop_delay` integer · Optional	Fan-stop delay after computer shutdown, in seconds. If omitted: Unchanged
`persist` boolean · Optional	true saves parameter-only changes; enablement changes are saved automatically. If omitted: false

Request example

JSON — Source: example-068.json

```
{"persist":true}
```

Response and completion

data.low_threshold_v/recovery_threshold_v/running/persisted.

Usage notes

Affects protection and automatic shutdown. Partial application can precede failure; read back configuration/state.

power.protection.status

Get voltage protection status

Methods: GET / POST
Path: `/api/v1/power/protection/status`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-069.json

```
{}
```

Response and completion

data.initialized/running/state/current_voltage_v/countdown_remaining_sec/recovery_timer_sec/protection_count/uptime_ms; devices has agx_powered/lpmu_powered/agx_connected.

Usage notes

initialized=false does not establish active protection.

power.stats

Get monitoring statistics

Methods: GET / POST
Path: `/api/v1/power/stats`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-070.json

```
{}
```

Response and completion

data.voltage_samples/power_chip_packets/crc_errors/timeout_errors/threshold_violations/avg_voltage_v/avg_current_a/avg_power_w/uptime_ms.

power.stats.reset

Reset monitoring statistics

Methods: POST / PUT / DELETE
Path: `/api/v1/power/stats/reset`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-071.json

```
{}
```

Response and completion

data.reset.

Usage notes

Resets statistics, not sensor calibration.

power.status

Get power monitoring status

Methods: GET / POST
Path: `/api/v1/power/status`
Minimum access: Public

Parameters

No business parameters.

See the shared data model.

Request example

JSON Source: example-072.json

```
{}
```

Response and completion

data.voltage; power_chip has voltage_v/current_a/power_w/valid/crc_valid; protection,stats,monitoring_active describe protection, aggregates, and sampling.

power.threshold.set

Set voltage thresholds

Methods: POST / PUT / DELETE
Path: `/api/v1/power/threshold/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`min_v` number · Required	Minimum voltage in V. If omitted: No default
`max_v` number · Required	Maximum voltage in V; greater than minimum. If omitted: No default

Request example

JSON Source: example-073.json

```
{"min_v":12,"max_v":24}
```

Response and completion

data.min_v/max_v.

Usage notes

Monitoring thresholds and protection policy are separate; use power.protection.set for protection.

power.voltage

Read current voltage

Methods: GET / POST
Path: `/api/v1/power/voltage`
Minimum access: Public

Parameters

Parameter	Meaning
`now` boolean · Optional	true samples now; false reads the latest value. If omitted: false

Request example

JSON Source: example-074.json

```
{}
```

Response and completion

data.voltage_v(V)/voltage_mv(mV)/adc_raw/timestamp_ms.

Computers and USB APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
device.fan.set	Set fan mode and duty
device.fan.status	Get fan status
device.ping	Test network connectivity (TCP port check)
device.power	Control device power (on/off/force_off)
device.power.status	Get power measurements
device.reset	Reset device
device.status	Get device power status
device.usb.set	Set USB MUX target
device.usb.status	Get USB MUX status

device.fan.set

Set fan mode and duty

Methods: POST / PUT / DELETE
Path: `/api/v1/device/fan/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`fan` integer · Required	Fan ID. If omitted: No default
`mode` string · Optional	off/manual/auto/curve; this compatibility entry has its own handling. If omitted: Unchanged
`duty` integer · Optional	Manual duty, 0-100 percent. If omitted: Unchanged

Request example

JSON Source: example-075.json

```
{"fan":0,"mode":"manual","duty":50}
```

Response and completion

data.fan/success.

Usage notes

Prefer separate fan.mode and fan.set calls for new integrations.

device.fan.status

Get fan status

Methods: GET / POST
Path: `/api/v1/device/fan/status`
Minimum access: Public

Parameters

Parameter	Meaning
`fan` integer · Optional	Fan ID. If omitted: All

Request example

JSON Source: example-076.json

```
{}
```

Response and completion

data.fans or one fan; compatibility temp follows this entry's model. Prefer fan.status for new integrations.

device.ping

Test network connectivity (TCP port check)

Methods: POST / PUT / DELETE
Path: `/api/v1/device/ping`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	Target IPv4 address or hostname. If omitted: No default
`timeout` integer · Optional	Request timeout in milliseconds. If omitted: 1000

Request example

JSON Source: example-077.json

```
{"host":"192.0.2.20","timeout":1000}
```

Response and completion

data.host/reachable; success may include latency_ms, failure error.

Usage notes

reachable=false is a diagnostic result, not proof the API did not execute.

device.power

Control device power (on/off/force_off)

Methods: POST / PUT / DELETE
Path: `/api/v1/device/power`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	Target computer: agx=inference, lpmu=application. If omitted: agx
`action` string · Required	on/off/toggle/force_off. If omitted: No default

Request example

JSON Source: example-078.json

```
{"device":"agx","action":"on"}
```

Response and completion

data.device/action/success reports the control call. Continue reading device.status.

Usage notes

force_off can lose unsaved data; prefer normal shutdown. Repeating toggle reverses state again.

device.power.status

Get power measurements

Methods: GET / POST
Path: `/api/v1/device/power/status`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-079.json

```
{}
```

Response and completion

data.rails entries: name/voltage_mv/current_ma/power_mw; total_power_mw. Units are mV/mA/mW.

device.reset

Reset device

Methods: POST / PUT / DELETE
Path: `/api/v1/device/reset`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	Target computer: agx=inference, lpmu=application. If omitted: agx

Request example

JSON Source: example-080.json

```
{"device":"agx"}
```

Response and completion

data.device/reset. Recheck computer and application state afterward.

Usage notes

Interrupts the target computer's work.

device.status

Get device power status

Methods: GET / POST
Path: `/api/v1/device/status`
Minimum access: Public

Parameters

Parameter	Meaning
`device` string · Optional	Target computer: agx=inference, lpmu=application. If omitted: agx

Request example

JSON Source: example-081.json

```
{"device":"agx"}
```

Response and completion

data.device/state/power_good/uptime_ms/boot_count; state is off/standby/on/booting/error.

Usage notes

Power state does not establish OS/application readiness.

device.usb.set

Set USB MUX target

Methods: POST / PUT / DELETE
Path: `/api/v1/device/usb/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`target` string · Required	esp32/agx/lpmu/disconnect. If omitted: No default

Request example

JSON Source: example-082.json

```
{"target":"agx"}
```

Response and completion

data.target/success.

Usage notes

Switching can interrupt control/transfers on the original route.

device.usb.status

Get USB MUX status

Methods: GET / POST
Path: `/api/v1/device/usb/status`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-083.json

```
{}
```

Response and completion

data.target/configured; target can be esp32/agx/lpmu/disconnected.

Network APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
network.eth.status	Get Ethernet status
network.hostname	Get/set hostname
network.status	Get overall network status
network.wifi.ap.config	Configure WiFi AP
network.wifi.ap.stations	Get connected AP stations
network.wifi.connect	Connect to WiFi network
network.wifi.disconnect	Disconnect from WiFi
network.wifi.mode	Get/set WiFi mode
network.wifi.scan	Scan for WiFi networks

network.eth.status

Get Ethernet status

Methods: GET / POST
Path: `/api/v1/network/eth/status`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-084.json

```
{}
```

Response and completion

data.status/link_up, config_error/config_error_name/interface_present, available IP/netmask/gateway/dns1/dns2, and transmit/receive stats.

network.hostname

Get/set hostname

Methods: POST / PUT / DELETE
Path: `/api/v1/network/hostname`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`hostname` string · Optional	New device hostname. If omitted: Query only

Request example

JSON Source: example-085.json

```
{}
```

Response and completion

data.hostname.

Usage notes

Can change access addresses and certificate name matching. Clients should check the actual address.

network.status

Get overall network status

Methods: GET / POST
Path: `/api/v1/network/status`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-086.json

```
{}
```

Response and completion

data.hostname/wifi_mode, ethernet, wifi_sta, and wifi_ap. ethernet also includes config_error/config_error_name/interface_present. IP and gateway fields depend on connection state.

network.wifi.ap.config

Configure WiFi AP

Methods: POST / PUT / DELETE
Path: `/api/v1/network/wifi/ap/config`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`ssid` string · Required	Access-point SSID. If omitted: No default
`password` string · Optional	Access-point password; omitted/empty creates an open AP. If omitted: Empty string
`channel` integer · Optional	Access-point channel. If omitted: Implementation initial value
`hidden` boolean · Optional	Hide the SSID. If omitted: false

Request example

JSON Source: example-087.json

```
{"ssid":"TianShanOS","password":"REPLACE_WITH_STRONG_PASSWORD","channel":6,"hidden":false}
```

Response and completion

data.ssid/auth/channel/hidden/configured.

Usage notes

Changing the hotspot can affect current connectivity.

network.wifi.ap.stations

Get connected AP stations

Methods: GET / POST
Path: `/api/v1/network/wifi/ap/stations`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-088.json

```
{}
```

Response and completion

data.count/stations with associated clients and available signal information.

network.wifi.connect

Connect to WiFi network

Methods: POST / PUT / DELETE
Path: `/api/v1/network/wifi/connect`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`ssid` string · Required	Target network SSID. If omitted: No default
`password` string · Optional	Network password. If omitted: Empty string

Request example

JSON Source: example-089.json

```
{"ssid":"YOUR_SSID","password":"YOUR_WIFI_PASSWORD"}
```

Response and completion

data.ssid/connecting. Query WiFi state until an address is acquired.

Usage notes

Can change the active network connection; prepare another access route.

network.wifi.disconnect

Disconnect from WiFi

Methods: POST / PUT / DELETE
Path: `/api/v1/network/wifi/disconnect`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-090.json

```
{}
```

Response and completion

data.disconnected. Management connectivity may also end.

network.wifi.mode

Get/set WiFi mode

Methods: POST / PUT / DELETE
Path: `/api/v1/network/wifi/mode`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`mode` string · Optional	off/sta/ap/apsta. If omitted: Query only

Request example

JSON Source: example-091.json

```
{"mode":"apsta"}
```

Response and completion

data.mode.

Usage notes

sta joins a network; ap offers a hotspot. Switching can disconnect management.

network.wifi.scan

Scan for WiFi networks

Methods: POST / PUT / DELETE
Path: `/api/v1/network/wifi/scan`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-092.json

```
{}
```

Response and completion

data.count/networks; entries ssid/rssi/channel/auth, plus bssid for wifi.scan. RSSI is dBm.

Usage notes

Starts a real scan. Both entries have the same access requirements but differing auth text/fields.

WiFi management APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
wifi.connect	Connect to WiFi network
wifi.disconnect	Disconnect from WiFi
wifi.scan	Scan for WiFi networks
wifi.status	Get WiFi status

wifi.connect

Connect to WiFi network

Methods: POST / PUT / DELETE
Path: `/api/v1/wifi/connect`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`ssid` string · Required	Target network SSID. If omitted: No default
`password` string · Optional	Network password. If omitted: Empty string

Request example

JSON Source: example-093.json

```
{"ssid":"YOUR_SSID","password":"YOUR_WIFI_PASSWORD"}
```

Response and completion

data.ssid/connecting. Query WiFi state until an address is acquired.

Usage notes

Can change the active network connection; prepare another access route.

wifi.disconnect

Disconnect from WiFi

Methods: POST / PUT / DELETE
Path: `/api/v1/wifi/disconnect`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-094.json

```
{}
```

Response and completion

data.disconnected. Management connectivity may also end.

wifi.scan

Scan for WiFi networks

Methods: POST / PUT / DELETE
Path: `/api/v1/wifi/scan`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-095.json

```
{}
```

Response and completion

data.count/networks; entries ssid/rssi/channel/auth, plus bssid for wifi.scan. RSSI is dBm.

Usage notes

Starts a real scan. Both entries have the same access requirements but differing auth text/fields.

wifi.status

Get WiFi status

Methods: GET / POST
Path: `/api/v1/wifi/status`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-096.json

```
{}
```

Response and completion

data.ap/sta with state/has_ip, available ip; STA may add gateway/rssi. Structure differs from network.status.

Address allocation APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
dhcp.clients	List DHCP clients
dhcp.start	Start DHCP server
dhcp.status	Get DHCP server status
dhcp.stop	Stop DHCP server

dhcp.clients

List DHCP clients

Methods: GET / POST
Path: `/api/v1/dhcp/clients`
Minimum access: READ

Parameters

Parameter	Meaning
`interface` string · Optional	ap/wifi or eth/ethernet. If omitted: Aggregate for queries; AP for start/stop

Request example

JSON Source: example-097.json

```
{"interface":"ap"}
```

Response and completion

data.clients/count; leases include mac/ip/hostname/lease_start/lease_expire/is_static.

dhcp.start

Start DHCP server

Methods: POST / PUT / DELETE
Path: `/api/v1/dhcp/start`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`interface` string · Optional	ap/wifi or eth/ethernet. If omitted: Aggregate for queries; AP for start/stop

Request example

JSON Source: example-098.json

```
{"interface":"ap"}
```

Response and completion

data.interface/started.

dhcp.status

Get DHCP server status

Methods: GET / POST
Path: `/api/v1/dhcp/status`
Minimum access: READ

Parameters

Parameter	Meaning
`interface` string · Optional	ap/wifi or eth/ethernet. If omitted: Aggregate for queries; AP for start/stop

Request example

JSON Source: example-099.json

```
{"interface":"ap"}
```

Response and completion

data.interfaces or one interface contains state/running, pool, gateway/dns/netmask/lease_min, and statistics. It also includes interface_present/restore_error/apply_error. eth adds want_running/link_up/resume_error; error code 0 means that step reported no error.

Usage notes

want_running is the requested state, not running. Check interface presence, link state, and recovery errors separately; obtaining an address alone is insufficient.

dhcp.stop

Stop DHCP server

Methods: POST / PUT / DELETE
Path: `/api/v1/dhcp/stop`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`interface` string · Optional	ap/wifi or eth/ethernet. If omitted: Aggregate for queries; AP for start/stop

Request example

JSON Source: example-100.json

```
{"interface":"ap"}
```

Response and completion

data.interface/stopped.

Network forwarding APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
nat.disable	Disable NAT gateway
nat.enable	Enable NAT gateway
nat.save	Save NAT configuration
nat.status	Get NAT gateway status

nat.disable

Disable NAT gateway

Methods: POST / PUT / DELETE
Path: `/api/v1/nat/disable`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-101.json

```
{}
```

Response and completion

data.success.

Usage notes

Enabling requires connected WiFi STA. Save runtime changes with nat.save.

nat.enable

Enable NAT gateway

Methods: POST / PUT / DELETE
Path: `/api/v1/nat/enable`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-102.json

```
{}
```

Response and completion

data.success.

Usage notes

Enabling requires connected WiFi STA. Save runtime changes with nat.save.

nat.save

Save NAT configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/nat/save`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-103.json

```
{}
```

Response and completion

data.success.

Usage notes

Enabling requires connected WiFi STA. Save runtime changes with nat.save.

nat.status

Get NAT gateway status

Methods: GET / POST
Path: `/api/v1/nat/status`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-104.json

```
{}
```

Response and completion

data.enabled/state/wifi_connected/eth_up.

Usage notes

Enabling requires connected WiFi STA. Save runtime changes with nat.save.

Time APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
time.force_sync	Force NTP sync
time.info	Get time sync info
time.set_ntp	Set NTP server
time.set_timezone	Set timezone
time.sync	Sync time from browser

time.force_sync

Force NTP sync

Methods: POST / PUT / DELETE
Path: `/api/v1/time/force_sync`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-105.json

```
{}
```

Response and completion

data.syncing/message. Inspect time.info for actual synchronization.

time.info

Get time sync info

Methods: GET / POST
Path: `/api/v1/time/info`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-106.json

```
{}
```

Response and completion

data.timestamp (Unix seconds), timestamp_ms, datetime/timezone, and source/sync_status/synced/last_sync/sync_count/ntp_server.

time.set_ntp

Set NTP server

Methods: POST / PUT / DELETE
Path: `/api/v1/time/set_ntp`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`server` string · Required	NTP server hostname or address. If omitted: No default
`index` integer · Optional	Server slot index. If omitted: 0

Request example

JSON Source: example-107.json

```
{"server":"YOUR_NTP_SERVER","index":0}
```

Response and completion

data.server/index/success.

time.set_timezone

Set timezone

Methods: POST / PUT / DELETE
Path: `/api/v1/time/set_timezone`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`timezone` string · Required	POSIX timezone string, such as CST-8. If omitted: No default

Request example

JSON Source: example-108.json

```
{"timezone":"CST-8"}
```

Response and completion

data.timezone/local_time/success.

time.sync

Sync time from browser

Methods: POST / PUT / DELETE
Path: `/api/v1/time/sync`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`timestamp_ms` integer · Required	Unix timestamp in milliseconds. If omitted: No default

Request example

JSON Source: example-109.json

```
{"timestamp_ms":1793894400000}
```

Response and completion

data.datetime/synced/source.

Usage notes

Changing system time affects certificate validity checks. Read time.info afterwards to confirm.

Device files APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
storage.delete	Delete file or directory
storage.info	Get file info
storage.list	List directory contents
storage.mkdir	Create directory
storage.mount	Mount SD card
storage.read	Read file content
storage.rename	Rename/move file
storage.status	Get storage status
storage.unmount	Unmount SD card

storage.delete

Delete file or directory

Methods: POST / PUT / DELETE
Path: `/api/v1/storage/delete`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`path` string · Required	Absolute device path; managed configuration and sensitive files have additional restrictions. If omitted: No default

Request example

JSON Source: example-110.json

```
{"path":"/sdcard/example.txt"}
```

Response and completion

data.deleted contains the deleted path.

Usage notes

Allows permitted paths under /sdcard/, /spiffs/, and /www/. Directories may be deleted recursively; managed configuration and storage roots remain protected.

storage.info

Get file info

Methods: GET / POST
Path: `/api/v1/storage/info`
Minimum access: READ

Parameters

Parameter	Meaning
`path` string · Required	Absolute device path; managed configuration and sensitive files have additional restrictions. If omitted: No default

Request example

JSON Source: example-111.json

```
{"path":"/sdcard/example.txt"}
```

Response and completion

data.path/name/type/size/modified. modified is filesystem Unix seconds.

storage.list

List directory contents

Methods: GET / POST
Path: `/api/v1/storage/list`
Minimum access: READ

Parameters

Parameter	Meaning
`path` string · Optional	Absolute directory path. If omitted: /sdcard
`recursive` boolean · Optional	Recursion is unimplemented; even true lists only this directory. Prefer omission. If omitted: false

Request example

JSON Source: example-112.json

```
{"path":"/sdcard"}
```

Response and completion

data.path/entries with name/type/size etc. Sensitive-directory listing requires ADMIN.

storage.mkdir

Create directory

Methods: POST / PUT / DELETE
Path: `/api/v1/storage/mkdir`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`path` string · Required	Absolute device path; managed configuration and sensitive files have additional restrictions. If omitted: No default

Request example

JSON Source: example-113.json

```
{"path":"/sdcard/example"}
```

Response and completion

data.created contains the directory path.

storage.mount

Mount SD card

Methods: POST / PUT / DELETE
Path: `/api/v1/storage/mount`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-114.json

```
{}
```

Response and completion

data.status; mount also includes path.

Usage notes

Affects operations using SD. End file/update work before unmounting.

storage.read

Read file content

Methods: GET / POST
Path: `/api/v1/storage/read`
Minimum access: READ

Parameters

Parameter	Meaning
`path` string · Required	Absolute device path; managed configuration and sensitive files have additional restrictions. If omitted: No default

Request example

JSON Source: example-115.json

```
{"path":"/sdcard/example.txt"}
```

Response and completion

data.path/size/type/content; content is parsed JSON for type=json, a string for type=text.

Usage notes

Maximum 8192-byte file. Sensitive paths require ADMIN. Use dedicated download for binary/large files.

storage.rename

Rename/move file

Methods: POST / PUT / DELETE
Path: `/api/v1/storage/rename`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`from` string · Required	Original absolute path. If omitted: No default
`to` string · Required	Destination absolute path. If omitted: No default

Request example

JSON Source: example-116.json

```
{"from":"/sdcard/example.txt","to":"/sdcard/renamed.txt"}
```

Response and completion

data.from/to.

Usage notes

Both paths must be within permitted /sdcard/, /spiffs/, or /www/ locations and pass managed-path checks. Filesystem and mount state can still limit moves.

storage.status

Get storage status

Methods: GET / POST
Path: `/api/v1/storage/status`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-117.json

```
{}
```

Response and completion

data.sd/spiffs with mounted and available path.

storage.unmount

Unmount SD card

Methods: POST / PUT / DELETE
Path: `/api/v1/storage/unmount`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-118.json

```
{}
```

Response and completion

data.status; mount also includes path.

Usage notes

Affects operations using SD. End file/update work before unmounting.

Controller services APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
service.list	List all services
service.restart	Restart a service
service.start	Start a service
service.status	Get service status
service.stop	Stop a service

service.list

List all services

Methods: GET / POST
Path: `/api/v1/service/list`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-119.json

```
{}
```

Response and completion

data.total/running/services; entries name/state/phase/healthy/start_time_ms/start_duration_ms.

service.restart

Restart a service

Methods: POST / PUT / DELETE
Path: `/api/v1/service/restart`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`name` string · Required	Service name obtained from service.list. If omitted: No default

Request example

JSON Source: example-120.json

```
{"name":"SERVICE_FROM_LIST"}
```

Response and completion

data.name/status.

Usage notes

Maintenance can interrupt network/WebUI/dependencies. Use automation.services for remote applications.

service.start

Start a service

Methods: POST / PUT / DELETE
Path: `/api/v1/service/start`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`name` string · Required	Service name obtained from service.list. If omitted: No default

Request example

JSON Source: example-121.json

```
{"name":"SERVICE_FROM_LIST"}
```

Response and completion

data.name/status.

Usage notes

Maintenance can interrupt network/WebUI/dependencies. Use automation.services for remote applications.

service.status

Get service status

Methods: GET / POST
Path: `/api/v1/service/status`
Minimum access: Public

Parameters

Parameter	Meaning
`name` string · Required	Service name obtained from service.list. If omitted: No default

Request example

JSON Source: example-122.json

```
{"name":"webui"}
```

Response and completion

data.name/state/phase/healthy/start_time_ms/start_duration_ms.

service.stop

Stop a service

Methods: POST / PUT / DELETE
Path: `/api/v1/service/stop`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`name` string · Required	Service name obtained from service.list. If omitted: No default

Request example

JSON Source: example-123.json

```
{"name":"SERVICE_FROM_LIST"}
```

Response and completion

data.name/status.

Usage notes

Maintenance can interrupt network/WebUI/dependencies. Use automation.services for remote applications.

Compute monitoring APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
monitor.config	Get Compute monitor configuration
monitor.data	Get latest Compute telemetry data
monitor.set_server	Set compute monitoring server IP and port
monitor.start	Start Compute monitoring
monitor.status	Get Compute monitor status
monitor.stop	Stop Compute monitoring

monitor.config

Get Compute monitor configuration

Methods: GET / POST
Path: `/api/v1/monitor/config`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-124.json

```
{}
```

Response and completion

data.server/port/reconnect_ms/heartbeat_timeout_ms/startup_delay_ms/auto_start/update_temp_source.

monitor.data

Get latest Compute telemetry data

Methods: GET / POST
Path: `/api/v1/monitor/data`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-125.json

```
{}
```

Response and completion

data.timestamp; cpu cores/data(id/usage/freq); memory ram/swap used/total(MB); temperature cpu/soc0/soc1/soc2/tj(Celsius); power gpu_soc/cpu_cv/sys_5v current/average(mW); gpu.gr3d_freq_pct.

Usage notes

Missing valid compute data returns a business error. These are compute-host metrics.

monitor.set_server

Set compute monitoring server IP and port

Methods: POST / PUT / DELETE
Path: `/api/v1/monitor/set_server`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`server` string · Required	Monitoring-server address. If omitted: No default
`port` integer · Optional	Port 1-65535; omission follows driver defaults. If omitted: 0 selects default handling

Request example

JSON Source: example-126.json

```
{"server":"192.0.2.20","port":58090}
```

Response and completion

data.server/port/saved.

monitor.start

Start Compute monitoring

Methods: POST / PUT / DELETE
Path: `/api/v1/monitor/start`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-127.json

```
{}
```

Response and completion

data.started.

monitor.status

Get Compute monitor status

Methods: GET / POST
Path: `/api/v1/monitor/status`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-128.json

```
{}
```

Response and completion

data.initialized/running/connection/reconnects/messages/errors/reliability/connected_time_ms; last_error may appear.

monitor.stop

Stop Compute monitoring

Methods: POST / PUT / DELETE
Path: `/api/v1/monitor/stop`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-129.json

```
{}
```

Response and completion

data.stopped.

Monitoring layout APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
ui.widgets.get	Get data widgets configuration
ui.widgets.set	Save data widgets configuration

ui.widgets.get

Get data widgets configuration

Methods: GET / POST
Path: `/api/v1/ui/widgets/get`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-130.json

```
{}
```

Response and completion

data.widgets/refresh_interval/source; layout is saved in device configuration.

ui.widgets.set

Save data widgets configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/ui/widgets/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`widgets` array · Required	Widget configuration array; see the widget model. If omitted: No default
`refresh_interval` integer · Optional	Refresh interval in milliseconds. If omitted: 5000

See the shared data model.

Request example

JSON Source: example-131.json

```
{"widgets":[],"refresh_interval":5000}
```

Response and completion

data.nvs_saved. Read layout back with ui.widgets.get.

Usage notes

Serialized configuration is limited to 4000 bytes.

Logs APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
log.capture	Enable/disable ESP_LOG capture
log.clear	Clear log buffer
log.list	Get log entries with filtering
log.setLevel	Set log level
log.stats	Get log system statistics

log.capture

Enable/disable ESP_LOG capture

Methods: POST / PUT / DELETE
Path: `/api/v1/log/capture`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`enable` boolean · Required	Capture system logs into the log buffer. If omitted: No default

Request example

JSON Source: example-132.json

```
{"enable":true}
```

Response and completion

data.capturing.

log.clear

Clear log buffer

Methods: POST / PUT / DELETE
Path: `/api/v1/log/clear`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-133.json

```
{}
```

Response and completion

data.cleared. Earlier in-memory logs can no longer be read from this buffer.

log.list

Get log entries with filtering

Methods: GET / POST
Path: `/api/v1/log/list`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`offset` integer · Optional	Starting offset within this search result. If omitted: 0
`limit` integer · Optional	Return limit, 1-200. If omitted: 50
`level` integer · Optional	Exact log level 1-5; do not combine with range filters. If omitted: All
`minLevel` integer · Optional	Minimum log level. If omitted: ERROR
`maxLevel` integer · Optional	Maximum log level. If omitted: VERBOSE
`tag` string · Optional	Module-name substring filter. If omitted: None
`keyword` string · Optional	Message keyword. If omitted: None

Request example

JSON Source: example-134.json

```
{"limit":50}
```

Response and completion

data.logs/total/offset/returned/bufferCapacity/bufferCount. Entries have timestamp(milliseconds since start),level/levelName/tag/message/task.

Usage notes

Memory history is bounded. offset applies within the selected search result, not an unlimited archive.

log.setLevel

Set log level

Methods: POST / PUT / DELETE
Path: `/api/v1/log/setLevel`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`level` integer / string · Required	Integer 0-5 or a level name such as error/warn/info/debug/verbose. If omitted: No default
`tag` string · Optional	Specific module; omit for global. If omitted: Global

Request example

JSON Source: example-135.json

```
{"level":3}
```

Response and completion

data.level is the level name; tag is included for a module-specific change.

log.stats

Get log system statistics

Methods: GET / POST
Path: `/api/v1/log/stats`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-136.json

```
{}
```

Response and completion

Log-buffer and capture statistics, indicating capacity/current recording state.

SSH fingerprints APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
hosts.clear	Clear all known hosts
hosts.info	Get known host info
hosts.list	List all known SSH hosts
hosts.remove	Remove a known host
hosts.update	Force update a host fingerprint

hosts.clear

Clear all known hosts

Methods: POST / PUT / DELETE
Path: `/api/v1/hosts/clear`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-137.json

```
{}
```

Response and completion

data.cleared.

Usage notes

Affects later SSH/SFTP trust checks; does not delete saved host connection definitions.

hosts.info

Get known host info

Methods: GET / POST
Path: `/api/v1/hosts/info`
Minimum access: READ

Parameters

Parameter	Meaning
`host` string · Required	Remote address; trust records match address and port. If omitted: No default
`port` integer · Optional	SSH port, 1-65535. If omitted: 22

Request example

JSON Source: example-138.json

```
{"host":"192.0.2.20","port":22}
```

Response and completion

data.host/port/type/fingerprint/added.

hosts.list

List all known SSH hosts

Methods: GET / POST
Path: `/api/v1/hosts/list`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-139.json

```
{}
```

Response and completion

data.hosts/count; entries host/port/type/fingerprint/added.

hosts.remove

Remove a known host

Methods: POST / PUT / DELETE
Path: `/api/v1/hosts/remove`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	Remote address; trust records match address and port. If omitted: No default
`port` integer · Optional	SSH port, 1-65535. If omitted: 22

Request example

JSON Source: example-140.json

```
{"host":"192.0.2.20"}
```

Response and completion

data.host/port/removed.

Usage notes

Removing trust does not revoke remote users/keys. Verify identity again before the next connection.

hosts.update

Force update a host fingerprint

Methods: POST / PUT / DELETE
Path: `/api/v1/hosts/update`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	Remote address; trust records match address and port. If omitted: No default
`port` integer · Optional	SSH port, 1-65535. If omitted: 22

Request example

JSON Source: example-141.json

```
{"host":"192.0.2.20"}
```

Response and completion

data.host/port/updated/message.

Usage notes

Connects to the remote host and updates trust. Independently verify its identity first.

SSH keys APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
key.delete	Delete a key
key.export	Export public key
key.exportPrivate	Export private key (only for exportable keys)
key.generate	Generate a new key
key.info	Get key info
key.list	List all stored keys

key.delete

Delete a key

Methods: POST / PUT / DELETE
Path: `/api/v1/key/delete`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Local key ID. If omitted: No default
`preview` boolean · Optional	true previews associated hosts/runs; false deletes. If omitted: false

Request example

JSON Source: example-142.json

```
{"id":"example-key","preview":true}
```

Response and completion

Preview: data.hosts/runs/complete. Delete: data.id/deleted.

Usage notes

Does not stop remote processes or remove public keys from authorized_keys. Deletion may remove the ability to verify/stop an existing run.

key.export

Export public key

Methods: GET / POST
Path: `/api/v1/key/export`
Minimum access: READ

Parameters

Parameter	Meaning
`id` string · Required	Local key ID. If omitted: No default

Request example

JSON Source: example-143.json

```
{"id":"example-key"}
```

Response and completion

data.id/type/public_key/comment; public_key is OpenSSH text.

Usage notes

Hidden-key reads require the appropriate role.

key.exportPrivate

Export private key (only for exportable keys)

Methods: POST / PUT / DELETE
Path: `/api/v1/key/exportPrivate`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Local key ID. If omitted: No default

Request example

JSON Source: example-144.json

```
{"id":"example-key"}
```

Response and completion

data.id/type/private_key/comment/warning.

Usage notes

Requires ADMIN and exportable=true; keys are non-exportable by default. Treat the response as secret, not ordinary log content.

key.generate

Generate a new key

Methods: POST / PUT / DELETE
Path: `/api/v1/key/generate`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Local key ID. If omitted: No default
`type` string · Optional	rsa2048/rsa4096/ec256/ecdsa/ec384. If omitted: rsa2048
`comment` string · Optional	Public-key comment. If omitted: Empty
`alias` string · Optional	Display alias. If omitted: Empty
`hidden` boolean · Optional	Hide the key. If omitted: false
`exportable` boolean · Optional	Permit later private-key export. If omitted: false
`request_id` string · Optional	Optional request correlation value, fewer than 32 bytes, alphanumeric/hyphen only. If omitted: None

Request example

JSON Source: example-145.json

```
{"id":"example-key","type":"ec256","exportable":false}
```

Response and completion

data.generated/id/type; failure may include failed_stage/esp_error/cleanup_complete/cleanup_error.

Usage notes

Check generated. Existing IDs and generation failures are errors; keys are not automatically overwritten.

key.info

Get key info

Methods: GET / POST
Path: `/api/v1/key/info`
Minimum access: READ

Parameters

Parameter	Meaning
`id` string · Required	Local key ID. If omitted: No default

Request example

JSON Source: example-146.json

```
{"id":"example-key"}
```

Response and completion

data.id/type/type_desc/comment/alias/created/last_used/has_public_key/exportable/hidden.

key.list

List all stored keys

Methods: GET / POST
Path: `/api/v1/key/list`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-147.json

```
{}
```

Response and completion

data.keys/count/max_keys; metadata id/type/type_desc/comment/alias/created/last_used/has_pubkey/exportable/hidden.

Usage notes

Below-ADMIN receives permitted key metadata only. Lists contain no private keys.

Device certificates APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
cert.delete	Delete all PKI credentials
cert.generate_csr	Generate Certificate Signing Request
cert.generate_keypair	Generate ECDSA P-256 key pair
cert.get_certificate	Get device certificate PEM
cert.install	Install signed certificate
cert.install_ca	Install CA certificate chain
cert.status	Read certificate and diagnostic-service status

cert.delete

Delete all PKI credentials

Methods: POST / PUT / DELETE
Path: `/api/v1/cert/delete`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`confirm` boolean · Required	Explicit true confirms deletion of local PKI material. If omitted: No default

Request example

JSON Source: example-148.json

```
{"confirm":true}
```

Response and completion

data.success/message.

Usage notes

Can change the separate HTTPS diagnostic-service identity or return it to an initial-certificate state. Does not revoke a certificate at its issuer.

cert.generate_csr

Generate Certificate Signing Request

Methods: POST / PUT / DELETE
Path: `/api/v1/cert/generate_csr`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device_id` string · Optional	Device identifier for the certificate subject. If omitted: Build-configured device ID
`organization` string · Optional	Organization name. If omitted: RMinte
`org_unit` string · Optional	Organizational unit. If omitted: Device

Request example

JSON Source: example-149.json

```
{"device_id":"YOUR_DEVICE_ID","organization":"YOUR_ORG","org_unit":"Device"}
```

Response and completion

data.success/csr_pem. Give the CSR to a trusted issuer; it contains no private key.

cert.generate_keypair

Generate ECDSA P-256 key pair

Methods: POST / PUT / DELETE
Path: `/api/v1/cert/generate_keypair`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`force` boolean · Optional	Request regeneration when identity material already exists. If omitted: false

Request example

JSON Source: example-150.json

```
{}
```

Response and completion

data.success/message.

Usage notes

Replacing the identity key can invalidate the existing certificate match. Plan maintenance first. This is not a key.* SSH key.

cert.get_certificate

Get device certificate PEM

Methods: GET / POST
Path: `/api/v1/cert/get_certificate`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-151.json

```
{}
```

Response and completion

data.cert_pem is the device certificate PEM, without private key.

cert.install

Install signed certificate

Methods: POST / PUT / DELETE
Path: `/api/v1/cert/install`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`cert_pem` string · Required	PEM certificate, at most 3999 content bytes; must match the existing device private key. If omitted: No default

Request example

JSON Source: example-152.json

```
{"cert_pem":"-----BEGIN CERTIFICATE-----\nYOUR_CERTIFICATE\n-----END CERTIFICATE-----"}
```

Response and completion

data.success/message/cert_info with subject_cn/issuer_cn/not_before/not_after/validity/time_ready and expiry information.

Usage notes

Check certificate storage, validity, and loading by the separate HTTPS diagnostic service independently. Business APIs remain on HTTP80. Read cert.status after installation.

cert.install_ca

Install CA certificate chain

Methods: POST / PUT / DELETE
Path: `/api/v1/cert/install_ca`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`ca_pem` string · Required	Trusted CA-chain PEM; parsing and capacity checks apply. If omitted: No default

Request example

JSON Source: example-153.json

```
{"ca_pem":"-----BEGIN CERTIFICATE-----\nYOUR_CA_CERTIFICATE\n-----END CERTIFICATE-----"}
```

Response and completion

data.success/message.

Usage notes

CA material affects trust. Install only administrator-verified chains.

cert.status

Read certificate and diagnostic-service status

Methods: GET / POST
Path: `/api/v1/cert/status`
Minimum access: Public

Parameters

No business parameters.

See the shared data model.

Request example

JSON Source: example-154.json

```
{}
```

Response and completion

data.has_private_key/has_certificate/has_ca_chain; certificate information, time_ready/validity/blocked_by, and the separate HTTPS diagnostic-service runtime are reported independently. See the certificate response model.

SSH connections and commands APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
ssh.cancel	Cancel running SSH exec session
ssh.commands.add	Add or update SSH command configuration
ssh.commands.export	Export SSH command configuration as .tscfg
ssh.commands.get	Get SSH command configuration by ID
ssh.commands.import	Import SSH command configuration from .tscfg
ssh.commands.list	List all SSH command configurations
ssh.commands.remove	Remove SSH command configuration
ssh.commands.update	Update an existing SSH commands configuration
ssh.copyid	Deploy public key to remote server
ssh.exec	Execute remote command via SSH
ssh.exec_status	Read owned current or latest SSH result
ssh.exec_stream	Execute remote command with streaming output via WebSocket
ssh.hosts.add	Add SSH host configuration
ssh.hosts.export	Export SSH host configuration as .tscfg
ssh.hosts.get	Get SSH host configuration by ID
ssh.hosts.import	Import SSH host configuration from .tscfg
ssh.hosts.list	List all SSH host configurations
ssh.hosts.remove	Remove SSH host configuration
ssh.hosts.update	Update an existing SSH hosts configuration
ssh.keygen	Generate SSH key pair
ssh.revoke	Revoke (remove) deployed public key from remote server
ssh.services.start	Start a stored managed service
ssh.test	Test SSH connection

ssh.cancel

Cancel running SSH exec session

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/cancel`
Minimum access: ROOT

Parameters

Parameter	Meaning
`session_id` integer · Required	Execution ID returned by ssh.exec_stream. If omitted: No default
`boot_id` string · Optional	When supplied with result_key, must match the start receipt. The initiating login session may omit both. If omitted: No default
`result_key` string · Optional	Private receipt key supplied together with boot_id. If omitted: No default

Request example

JSON Source: example-155.json

```
{"session_id":1}
```

Response and completion

data.cancelled=true accepts cancellation intent; continue querying final status.

Usage notes

Requires the initiating login session; another root session cannot cancel it. A supplied receipt is still checked as a complete matching pair. cancelled acknowledges intent only; exec_status still needs the complete receipt.

ssh.commands.add

Add or update SSH command configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/commands/add`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-156.json

```
{
  "id": "example-command",
  "host_id": "example-host",
  "name": "Read OS",
  "command": "uname -a",
  "timeout_sec": 30,
  "enabled": true,
  "service_mode": false,
  "active_version": "CURRENT_VERSION"
}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Add requires an unused ID; update requires an existing object. See complete model examples.

ssh.commands.export

Export SSH command configuration as .tscfg

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/commands/export`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`recipient_cert` string · Optional	Recipient certificate PEM; omission encrypts for this device. If omitted: This device certificate
`include_host` boolean · Optional	Include the dependent host definition in export. If omitted: false

See the shared data model.

Request example

JSON Source: example-157.json

```
{"id":"example"}
```

Response and completion

data.tscfg/filename. Export additionally depends on Developer device identity and recipient materials.

Usage notes

The client saves returned package text. It does not become another device's active configuration automatically.

ssh.commands.get

Get SSH command configuration by ID

Methods: GET / POST
Path: `/api/v1/ssh/commands/get`
Minimum access: READ

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default

See the shared data model.

Request example

JSON Source: example-158.json

```
{"id":"example"}
```

Response and completion

One SSH command and active_version; below-ADMIN fields are filtered.

ssh.commands.import

Import SSH command configuration from .tscfg

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/commands/import`
Minimum access: ROOT

Parameters

Parameter	Meaning
`tscfg` string · Required	Complete package JSON text supplied as a string. If omitted: No default
`preview` boolean · Optional	Preview only when true. If omitted: false
`active_version` string · Required for apply	Current configuration version. If omitted: Preview may use current version
`overwrite` boolean · Optional	Permit replacing an existing object with different content. If omitted: false
`expected_credential` integer · Required for apply	credential_generation returned by preview. If omitted: None
`expected_digest` string · Required for apply	package_digest returned by preview. If omitted: None

See the shared data model.

Request example

JSON Source: example-159.json

```
{"tscfg":"PACKAGE_JSON_TEXT","preview":true}
```

Response and completion

Configuration-result model. Preview adds valid/items/exists/unchanged/package_digest/credential_generation; apply adds imported/overwritten/restart_required.

Usage notes

Preview first, then apply the same package with digest,credential generation,and configuration version. Preview again if conditions change.

ssh.commands.list

List all SSH command configurations

Methods: GET / POST
Path: `/api/v1/ssh/commands/list`
Minimum access: READ

Parameters

Parameter	Meaning
`offset` integer · Optional	Nonnegative return offset. If omitted: 0
`limit` integer · Optional	Return count; omission does not request truncation. If omitted: All
`host_id` string · Optional	Filter by saved host ID. If omitted: All

See the shared data model.

Request example

JSON Source: example-160.json

```
{}
```

Response and completion

data.commands/count/total/active_version/load_error/paused; below-ADMIN gets restricted metadata.

ssh.commands.remove

Remove SSH command configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/commands/remove`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`preview` boolean · Optional	true previews impact; false performs deletion. If omitted: false
`replacement_id` string · Optional	Replacement object ID for dependent references. If omitted: None
`mode` string · Optional	cascade requests cascade handling; choose one strategy rather than combining force/replacement. If omitted: Ordinary deletion
`force` boolean · Optional	Force deletion of a referenced object only after previewing impact. If omitted: false
`pause_automation` boolean · Optional	Pause automation before handling. If omitted: false

See the shared data model.

Request example

JSON Source: example-161.json

```
{"id":"example","active_version":"CURRENT_VERSION","preview":true}
```

Response and completion

Configuration-result model plus impact describing references affected.

Usage notes

Deleting local definitions does not automatically stop remote programs or revoke public keys.

ssh.commands.update

Update an existing SSH commands configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/commands/update`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-162.json

```
{
  "id": "example-command",
  "host_id": "example-host",
  "name": "Read OS",
  "command": "uname -a",
  "timeout_sec": 30,
  "enabled": true,
  "service_mode": false,
  "active_version": "CURRENT_VERSION"
}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Add requires an unused ID; update requires an existing object. See complete model examples.

ssh.copyid

Deploy public key to remote server

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/copyid`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	Remote address. If omitted: No default
`user` string · Required	Remote account. If omitted: No default
`port` integer · Optional	SSH port. If omitted: 22
`password` string · Required	Password for this remote authentication. If omitted: No default
`keyid` string · Required	Local key ID whose public key is being deployed/revoked. If omitted: No default
`trust_new` boolean · Optional	Accept a new host only after independent verification. If omitted: false
`accept_changed` boolean · Optional	Accept a changed host fingerprint only after verification. If omitted: false
`confirmed_fingerprint` string · Optional	Verified fingerprint matching the host being confirmed. If omitted: None
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`verify` boolean · Optional	Verify after deployment. If omitted: true

Request example

JSON Source: example-163.json

```
{
  "host": "192.0.2.20",
  "user": "YOUR_USER",
  "password": "REMOTE_PASSWORD",
  "keyid": "example-key",
  "active_version": "CURRENT_VERSION"
}
```

Response and completion

data.deployed/verified/registered/registration_error,host_id and host/key details. Check remote deployment separately from local registration.

Usage notes

Remote deployment can succeed while local registration fails. Do not immediately install again.

ssh.exec

Execute remote command via SSH

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/exec`
Minimum access: ROOT

Parameters

Parameter	Meaning
`host` string · Required	SSH host address. If omitted: No default
`user` string · Required	Remote username. If omitted: No default
`port` integer · Optional	Port 1-65535. If omitted: 22
`password` string · Authentication alternative	Password for this remote login, not the TianshanOS account. If omitted: None
`keyid` string · Authentication alternative	Saved SSH private-key ID on the device. If omitted: None
`keypath` string · Supported immediate-call authentication alternative	Device private-key path; support varies by entry point. If omitted: None
`command` string · Required	Remote command text. If omitted: No default
`timeout_ms` integer · Optional	Connection/execution deadline in milliseconds. If omitted: SSH default
`trust_new` boolean · Optional	Accept a new host only after independent verification. If omitted: false
`accept_changed` boolean · Optional	Accept a changed host fingerprint only after verification. If omitted: false
`confirmed_fingerprint` string · Optional	Verified fingerprint matching the host being confirmed. If omitted: None

Request example

JSON Source: example-164.json

```
{"host":"192.0.2.20","user":"YOUR_USER","keyid":"example-key","command":"uname -a"}
```

Response and completion

data.stdout/stderr/exit_code/host_status/fingerprint. Check exit code and application output.

Usage notes

Unknown/changed fingerprints return confirmation information. Verify before accepting. A nonzero remote exit code need not make top-level code nonzero.

ssh.exec_status

Read owned current or latest SSH result

Methods: GET / POST
Path: `/api/v1/ssh/exec_status`
Minimum access: ROOT

Parameters

Parameter	Meaning
`session_id` integer · Required	Execution ID returned by ssh.exec_stream. If omitted: No default
`boot_id` string · Required	32-character hexadecimal boot identity from the receipt. If omitted: No default
`result_key` string · Required	32-character hexadecimal result credential from the receipt. If omitted: No default

Request example

JSON Source: example-165.json

```
{
  "session_id": 1,
  "boot_id": "00000000000000000000000000000000",
  "result_key": "00000000000000000000000000000000"
}
```

Response and completion

data uses WebSocket result format: ssh_exec_running while active, ssh_exec_done/cancelled/error at termination. Check status/exit_code/success and output completeness.

Usage notes

Requires the initiating session and correct receipt. Zero-filled sample values show shape only; use the actual start response. Querying does not rerun the command.

ssh.exec_stream

Execute remote command with streaming output via WebSocket

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/exec_stream`
Minimum access: ROOT

Parameters

Parameter	Meaning
`host` string · Required	SSH host address. If omitted: No default
`user` string · Required	Remote username. If omitted: No default
`port` integer · Optional	Port 1-65535. If omitted: 22
`password` string · Authentication alternative	Password for this remote login, not the TianshanOS account. If omitted: None
`keyid` string · Authentication alternative	Saved SSH private-key ID on the device. If omitted: None
`command` string · Required	Command text. If omitted: No default
`expect_pattern` string · Optional	Success-match pattern. If omitted: Unchanged
`fail_pattern` string · Optional	Failure-match pattern. If omitted: Unchanged
`extract_pattern` string · Optional	Output-extraction pattern. If omitted: Unchanged
`var_name` string · Optional	Result-variable prefix. If omitted: Unchanged
`timeout` integer · Optional	Milliseconds here, unlike saved-command timeout_sec. If omitted: Default execution deadline
`collect_output` boolean · Optional	Retain collected output. If omitted: true
`max_output_size` integer · Optional	Output-collection limit in bytes. If omitted: Device default
`stop_on_match` boolean · Optional	Stop after a match. If omitted: false
`command_id` string · Optional pair with command_instance	Associated saved-command ID. If omitted: None
`command_instance` string · Optional pair with command_id	16-character lowercase hexadecimal identity matching runtime_instance from the list. If omitted: None

Request example

JSON Source: example-166.json

```
{
  "host": "192.0.2.20",
  "user": "YOUR_USER",
  "keyid": "example-key",
  "command": "uname -a",
  "collect_output": true
}
```

Response and completion

data.session_id/boot_id/result_key. Preserve the receipt before consuming WebSocket results or querying ssh.exec_status.

Usage notes

Exclusive streaming slot. No generic keypath authentication here. Verify ownership,terminal status,and output completeness.

ssh.hosts.add

Add SSH host configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/hosts/add`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-167.json

```
{
  "id": "example-host",
  "host": "192.0.2.20",
  "port": 22,
  "username": "YOUR_USER",
  "auth_type": "key",
  "keyid": "example-key",
  "enabled": true,
  "active_version": "CURRENT_VERSION"
}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Add requires an unused ID; update requires an existing object. See complete model examples.

ssh.hosts.export

Export SSH host configuration as .tscfg

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/hosts/export`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`recipient_cert` string · Optional	Recipient certificate PEM; omission encrypts for this device. If omitted: This device certificate

See the shared data model.

Request example

JSON Source: example-168.json

```
{"id":"example"}
```

Response and completion

data.tscfg/filename. Export additionally depends on Developer device identity and recipient materials.

Usage notes

The client saves returned package text. It does not become another device's active configuration automatically.

ssh.hosts.get

Get SSH host configuration by ID

Methods: GET / POST
Path: `/api/v1/ssh/hosts/get`
Minimum access: READ

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default

See the shared data model.

Request example

JSON Source: example-169.json

```
{"id":"example"}
```

Response and completion

One host and active_version; below-ADMIN fields are filtered.

ssh.hosts.import

Import SSH host configuration from .tscfg

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/hosts/import`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`tscfg` string · Required	Complete package JSON text supplied as a string. If omitted: No default
`preview` boolean · Optional	Preview only when true. If omitted: false
`active_version` string · Required for apply	Current configuration version. If omitted: Preview may use current version
`overwrite` boolean · Optional	Permit replacing an existing object with different content. If omitted: false
`expected_credential` integer · Required for apply	credential_generation returned by preview. If omitted: None
`expected_digest` string · Required for apply	package_digest returned by preview. If omitted: None

See the shared data model.

Request example

JSON Source: example-170.json

```
{"tscfg":"PACKAGE_JSON_TEXT","preview":true}
```

Response and completion

Configuration-result model. Preview adds valid/items/exists/unchanged/package_digest/credential_generation; apply adds imported/overwritten/restart_required.

Usage notes

Preview first, then apply the same package with digest,credential generation,and configuration version. Preview again if conditions change.

ssh.hosts.list

List all SSH host configurations

Methods: GET / POST
Path: `/api/v1/ssh/hosts/list`
Minimum access: READ

Parameters

Parameter	Meaning
`offset` integer · Optional	Nonnegative return offset. If omitted: 0
`limit` integer · Optional	Return count; omission does not request truncation. If omitted: All

See the shared data model.

Request example

JSON Source: example-171.json

```
{}
```

Response and completion

data.hosts/count/total/active_version/load_error/paused; below-ADMIN gets restricted metadata.

ssh.hosts.remove

Remove SSH host configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/hosts/remove`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`preview` boolean · Optional	true previews impact; false performs deletion. If omitted: false
`replacement_id` string · Optional	Replacement object ID for dependent references. If omitted: None
`mode` string · Optional	cascade requests cascade handling; choose one strategy rather than combining force/replacement. If omitted: Ordinary deletion
`force` boolean · Optional	Force deletion of a referenced object only after previewing impact. If omitted: false
`pause_automation` boolean · Optional	Pause automation before handling. If omitted: false

See the shared data model.

Request example

JSON Source: example-172.json

```
{"id":"example","active_version":"CURRENT_VERSION","preview":true}
```

Response and completion

Configuration-result model plus impact describing references affected.

Usage notes

Deleting local definitions does not automatically stop remote programs or revoke public keys.

ssh.hosts.update

Update an existing SSH hosts configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/hosts/update`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-173.json

```
{
  "id": "example-host",
  "host": "192.0.2.20",
  "port": 22,
  "username": "YOUR_USER",
  "auth_type": "key",
  "keyid": "example-key",
  "enabled": true,
  "active_version": "CURRENT_VERSION"
}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Add requires an unused ID; update requires an existing object. See complete model examples.

ssh.keygen

Generate SSH key pair

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/keygen`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Key ID. If omitted: No default
`type` string · Optional	Key type accepted here, such as rsa2048/rsa4096/ec256/ec384. If omitted: rsa2048
`comment` string · Optional	Public-key comment. If omitted: Empty

Request example

JSON Source: example-174.json

```
{"id":"example-key","type":"ec256"}
```

Response and completion

data.id/type/public_key.

Usage notes

Prefer key.generate for new integrations; supported parameters differ.

ssh.revoke

Revoke (remove) deployed public key from remote server

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/revoke`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	Remote address. If omitted: No default
`user` string · Required	Remote account. If omitted: No default
`port` integer · Optional	SSH port. If omitted: 22
`password` string · Required	Password for this remote authentication. If omitted: No default
`keyid` string · Required	Local key ID whose public key is being deployed/revoked. If omitted: No default
`trust_new` boolean · Optional	Accept a new host only after independent verification. If omitted: false
`accept_changed` boolean · Optional	Accept a changed host fingerprint only after verification. If omitted: false
`confirmed_fingerprint` string · Optional	Verified fingerprint matching the host being confirmed. If omitted: None

Request example

JSON Source: example-175.json

```
{"host":"192.0.2.20","user":"YOUR_USER","password":"REMOTE_PASSWORD","keyid":"example-key"}
```

Response and completion

data.revoked/found/removed_count and remote details.

Usage notes

Removes the specified public key only, not local keys,host definitions,or remote processes.

ssh.services.start

Start a stored managed service

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/services/start`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`command_id` string · Required	Saved command ID configured as a managed service. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

Request example

JSON Source: example-176.json

```
{"command_id":"example-service","active_version":"CURRENT_VERSION"}
```

Response and completion

data.operation_id/run_id; verify with automation.services.status.

Usage notes

Uses saved configuration. Distinguish process running from application ready even after call success.

ssh.test

Test SSH connection

Methods: POST / PUT / DELETE
Path: `/api/v1/ssh/test`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	SSH host address. If omitted: No default
`user` string · Required	Remote username. If omitted: No default
`port` integer · Optional	Port 1-65535. If omitted: 22
`password` string · Authentication alternative	Password for this remote login, not the TianshanOS account. If omitted: None
`keyid` string · Authentication alternative	Saved SSH private-key ID on the device. If omitted: None
`keypath` string · Supported immediate-call authentication alternative	Device private-key path; support varies by entry point. If omitted: None
`trust_new` boolean · Optional	Accept a new host only after independent verification. If omitted: false
`accept_changed` boolean · Optional	Accept a changed host fingerprint only after verification. If omitted: false
`confirmed_fingerprint` string · Optional	Verified fingerprint matching the host being confirmed. If omitted: None

Request example

JSON Source: example-177.json

```
{"host":"192.0.2.20","user":"YOUR_USER","keyid":"example-key"}
```

Response and completion

data.success/host/port/user/host_status/fingerprint; connection failure may include stage/reason/error and fingerprints.

Usage notes

Establishes connection/authentication conditions, not the usability of every remote command/service.

Remote files APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
sftp.get	Download file via SFTP
sftp.ls	List remote directory via SFTP
sftp.mkdir	Create remote directory via SFTP
sftp.put	Upload file via SFTP
sftp.rm	Delete remote file via SFTP
sftp.stat	Get remote file information via SFTP

sftp.get

Download file via SFTP

Methods: POST / PUT / DELETE
Path: `/api/v1/sftp/get`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	SSH host address. If omitted: No default
`user` string · Required	Remote username. If omitted: No default
`port` integer · Optional	Port 1-65535. If omitted: 22
`password` string · Authentication alternative	Password for this remote login, not the TianshanOS account. If omitted: None
`keyid` string · Authentication alternative	Saved SSH private-key ID on the device. If omitted: None
`keypath` string · Supported immediate-call authentication alternative	Device private-key path; support varies by entry point. If omitted: None
`remote` string · Required	Absolute remote file path. If omitted: No default
`local` string · Required	Absolute local device file path. If omitted: No default

Request example

JSON Source: example-178.json

```
{
  "host": "192.0.2.20",
  "user": "YOUR_USER",
  "keyid": "example-key",
  "remote": "/tmp/example.txt",
  "local": "/sdcard/example.txt"
}
```

Response and completion

data.remote/local/success; writes a local device file.

Usage notes

Requires verified remote host trust and authentication. Failures do not automatically accept fingerprints. get has local write restrictions; put sends local content remotely.

sftp.ls

List remote directory via SFTP

Methods: POST / PUT / DELETE
Path: `/api/v1/sftp/ls`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	SSH host address. If omitted: No default
`user` string · Required	Remote username. If omitted: No default
`port` integer · Optional	Port 1-65535. If omitted: 22
`password` string · Authentication alternative	Password for this remote login, not the TianshanOS account. If omitted: None
`keyid` string · Authentication alternative	Saved SSH private-key ID on the device. If omitted: None
`keypath` string · Supported immediate-call authentication alternative	Device private-key path; support varies by entry point. If omitted: None
`path` string · Required	Remote directory/file path. If omitted: No default

Request example

JSON Source: example-179.json

```
{"host":"192.0.2.20","user":"YOUR_USER","keyid":"example-key","path":"/tmp"}
```

Response and completion

data.path/files; entries name/size/is_dir/mtime/permissions.

Usage notes

Requires verified remote host trust and authentication. Failures do not automatically accept fingerprints. get has local write restrictions; put sends local content remotely.

sftp.mkdir

Create remote directory via SFTP

Methods: POST / PUT / DELETE
Path: `/api/v1/sftp/mkdir`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	SSH host address. If omitted: No default
`user` string · Required	Remote username. If omitted: No default
`port` integer · Optional	Port 1-65535. If omitted: 22
`password` string · Authentication alternative	Password for this remote login, not the TianshanOS account. If omitted: None
`keyid` string · Authentication alternative	Saved SSH private-key ID on the device. If omitted: None
`keypath` string · Supported immediate-call authentication alternative	Device private-key path; support varies by entry point. If omitted: None
`path` string · Required	Remote directory/file path. If omitted: No default
`mode` integer · Optional	Permission bits as a decimal JSON integer; octal 0755 is decimal 493. If omitted: 493

Request example

JSON Source: example-180.json

```
{"host":"192.0.2.20","user":"YOUR_USER","keyid":"example-key","path":"/tmp"}
```

Response and completion

data.path/created.

Usage notes

Requires verified remote host trust and authentication. Failures do not automatically accept fingerprints. get has local write restrictions; put sends local content remotely.

sftp.put

Upload file via SFTP

Methods: POST / PUT / DELETE
Path: `/api/v1/sftp/put`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	SSH host address. If omitted: No default
`user` string · Required	Remote username. If omitted: No default
`port` integer · Optional	Port 1-65535. If omitted: 22
`password` string · Authentication alternative	Password for this remote login, not the TianshanOS account. If omitted: None
`keyid` string · Authentication alternative	Saved SSH private-key ID on the device. If omitted: None
`keypath` string · Supported immediate-call authentication alternative	Device private-key path; support varies by entry point. If omitted: None
`remote` string · Required	Absolute remote file path. If omitted: No default
`local` string · Required	Absolute local device file path. If omitted: No default

Request example

JSON Source: example-181.json

```
{
  "host": "192.0.2.20",
  "user": "YOUR_USER",
  "keyid": "example-key",
  "remote": "/tmp/example.txt",
  "local": "/sdcard/example.txt"
}
```

Response and completion

data.remote/local/success; writes a remote file.

Usage notes

Requires verified remote host trust and authentication. Failures do not automatically accept fingerprints. get has local write restrictions; put sends local content remotely.

sftp.rm

Delete remote file via SFTP

Methods: POST / PUT / DELETE
Path: `/api/v1/sftp/rm`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	SSH host address. If omitted: No default
`user` string · Required	Remote username. If omitted: No default
`port` integer · Optional	Port 1-65535. If omitted: 22
`password` string · Authentication alternative	Password for this remote login, not the TianshanOS account. If omitted: None
`keyid` string · Authentication alternative	Saved SSH private-key ID on the device. If omitted: None
`keypath` string · Supported immediate-call authentication alternative	Device private-key path; support varies by entry point. If omitted: None
`path` string · Required	Remote directory/file path. If omitted: No default

Request example

JSON Source: example-182.json

```
{"host":"192.0.2.20","user":"YOUR_USER","keyid":"example-key","path":"/tmp"}
```

Response and completion

data.path/deleted.

Usage notes

Requires verified remote host trust and authentication. Failures do not automatically accept fingerprints. get has local write restrictions; put sends local content remotely.

sftp.stat

Get remote file information via SFTP

Methods: POST / PUT / DELETE
Path: `/api/v1/sftp/stat`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`host` string · Required	SSH host address. If omitted: No default
`user` string · Required	Remote username. If omitted: No default
`port` integer · Optional	Port 1-65535. If omitted: 22
`password` string · Authentication alternative	Password for this remote login, not the TianshanOS account. If omitted: None
`keyid` string · Authentication alternative	Saved SSH private-key ID on the device. If omitted: None
`keypath` string · Supported immediate-call authentication alternative	Device private-key path; support varies by entry point. If omitted: None
`path` string · Required	Remote directory/file path. If omitted: No default

Request example

JSON Source: example-183.json

```
{"host":"192.0.2.20","user":"YOUR_USER","keyid":"example-key","path":"/tmp"}
```

Response and completion

data.path/size/is_dir/is_link/uid/gid/permissions/atime/mtime.

Usage notes

Requires verified remote host trust and authentication. Failures do not automatically accept fingerprints. get has local write restrictions; put sends local content remotely.

Automation and managed services APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
automation.action.stats	Get action execution statistics
automation.action.stats.reset	Reset action statistics
automation.actions.add	Create a new action template
automation.actions.delete	Delete an action template
automation.actions.execute	Execute an action template
automation.actions.export	Export action template as config pack (.tscfg)
automation.actions.get	Get action template by ID
automation.actions.import	Import action template from config pack
automation.actions.list	List all action templates
automation.actions.update	Update an existing action template
automation.config.apply	Apply the confirmed restore or source choice
automation.config.backup	Save the complete emergency configuration
automation.config.enable	Explicitly enable or pause the current configuration
automation.config.preview	Preview a configuration restore or source choice
automation.config.refresh	Check SD card once without changing the user choice
automation.config.status	Current automation storage and backup status
automation.pause	Pause automation engine
automation.proxy.fetch	Proxy fetch from external REST API
automation.proxy.socketio_test	Test Socket.IO connection (v4 protocol)
automation.proxy.websocket_test	Test WebSocket connection
automation.reload	Reload configuration
automation.resume	Resume automation engine
automation.rules.add	Add a new rule
automation.rules.delete	Delete a rule

API	Purpose
automation.rules.disable	Disable a rule
automation.rules.enable	Enable a rule
automation.rules.export	Export rule as config pack (.tscfg)
automation.rules.get	Get rule details by ID
automation.rules.import	Import rule from config pack
automation.rules.list	List all rules
automation.rules.trigger	Manually trigger a rule
automation.rules.update	Update rule with expected revision
automation.services.list	List independent remote service controls
automation.services.status	Confirm managed remote process state
automation.services.stop	Stop verified managed remote process
automation.sources.add	Add a new data source
automation.sources.add_mapping	Add a field mapping to an existing data source
automation.sources.delete	Delete a data source
automation.sources.disable	Disable a data source
automation.sources.enable	Enable a data source
automation.sources.export	Export data source as config pack (.tscfg)
automation.sources.import	Import data source from config pack
automation.sources.list	List all data sources
automation.sources.remove_mapping	Remove a field mapping from a data source
automation.sources.update	Update an existing data source
automation.start	Start automation engine
automation.status	Get automation engine status
automation.stop	Stop automation engine
automation.test.device	Test device control action
automation.test.gpio	Test GPIO action
automation.test.led	Test LED action
automation.test.ssh	Test SSH command execution

API	Purpose
automation.variables.delete	Delete a saved user variable
automation.variables.get	Get variable value
automation.variables.list	List all variables
automation.variables.set	Set variable value

automation.action.stats

Get action execution statistics

Methods: GET / POST
Path: `/api/v1/automation/action/stats`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-184.json

```
{}
```

Response and completion

Current action-manager runtime statistics, not a persistent audit trail.

automation.action.stats.reset

Reset action statistics

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/action/stats/reset`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-185.json

```
{}
```

Response and completion

code=0 reports the reset request succeeded.

automation.actions.add

Create a new action template

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/actions/add`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-186.json

```
{
  "id": "example-action",
  "name": "Record a message",
  "enabled": true,
  "type": "log",
  "log": {
    "level": 3,
    "message": "Example action"
  },
  "active_version": "CURRENT_VERSION"
}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Add requires an unused ID; update requires an existing object. See complete model examples.

automation.actions.delete

Delete an action template

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/actions/delete`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`preview` boolean · Optional	true previews impact; false performs deletion. If omitted: false
`replacement_id` string · Optional	Replacement object ID for dependent references. If omitted: None
`mode` string · Optional	cascade requests cascade handling; choose one strategy rather than combining force/replacement. If omitted: Ordinary deletion
`force` boolean · Optional	Force deletion of a referenced object only after previewing impact. If omitted: false
`pause_automation` boolean · Optional	Pause automation before handling. If omitted: false

See the shared data model.

Request example

JSON Source: example-187.json

```
{"id":"example","active_version":"CURRENT_VERSION","preview":true}
```

Response and completion

Configuration-result model plus impact describing references affected.

Usage notes

Deleting local definitions does not automatically stop remote programs or revoke public keys.

automation.actions.execute

Execute an action template

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/actions/execute`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-188.json

```
{"id":"example-action","active_version":"CURRENT_VERSION"}
```

Response and completion

data.async/status/duration_ms and execution results; private output is filtered for below-ADMIN.

Usage notes

Executes the template's real hardware/remote effects. Continue checking the target result for asynchronous templates.

automation.actions.export

Export action template as config pack (.tscfg)

Methods: POST
Path: `/api/v1/automation/actions/export`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`recipient_cert` string · Optional	Recipient certificate PEM; omission encrypts for this device. If omitted: This device certificate

See the shared data model.

Request example

JSON Source: example-189.json

```
{"id":"example"}
```

Response and completion

data.tscfg/filename. Export additionally depends on Developer device identity and recipient materials.

Usage notes

The client saves returned package text. It does not become another device's active configuration automatically.

automation.actions.get

Get action template by ID

Methods: GET / POST
Path: `/api/v1/automation/actions/get`
Minimum access: READ

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default

See the shared data model.

Request example

JSON Source: example-190.json

```
{"id":"example"}
```

Response and completion

One action template and active_version; below-ADMIN fields are filtered.

automation.actions.import

Import action template from config pack

Methods: POST
Path: `/api/v1/automation/actions/import`
Minimum access: ROOT

Parameters

Parameter	Meaning
`tscfg` string · Required	Complete package JSON text supplied as a string. If omitted: No default
`preview` boolean · Optional	Preview only when true. If omitted: false
`active_version` string · Required for apply	Current configuration version. If omitted: Preview may use current version
`overwrite` boolean · Optional	Permit replacing an existing object with different content. If omitted: false
`expected_credential` integer · Required for apply	credential_generation returned by preview. If omitted: None
`expected_digest` string · Required for apply	package_digest returned by preview. If omitted: None

See the shared data model.

Request example

JSON Source: example-191.json

```
{"tscfg":"PACKAGE_JSON_TEXT","preview":true}
```

Response and completion

Configuration-result model. Preview adds valid/items/exists/unchanged/package_digest/credential_generation; apply adds imported/overwritten/restart_required.

Usage notes

Preview first, then apply the same package with digest,credential generation,and configuration version. Preview again if conditions change.

automation.actions.list

List all action templates

Methods: GET / POST
Path: `/api/v1/automation/actions/list`
Minimum access: READ

Parameters

Parameter	Meaning
`offset` integer · Optional	Nonnegative return offset. If omitted: 0
`limit` integer · Optional	Return count; omission does not request truncation. If omitted: All

See the shared data model.

Request example

JSON Source: example-192.json

```
{}
```

Response and completion

data.templates/count/total/active_version/load_error/paused; below-ADMIN gets restricted metadata.

automation.actions.update

Update an existing action template

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/actions/update`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-193.json

```
{
  "id": "example-action",
  "name": "Record a message",
  "enabled": true,
  "type": "log",
  "log": {
    "level": 3,
    "message": "Example action"
  },
  "active_version": "CURRENT_VERSION"
}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Add requires an unused ID; update requires an existing object. See complete model examples.

automation.config.apply

Apply the confirmed restore or source choice

Methods: POST
Path: `/api/v1/automation/config/apply`
Minimum access: ROOT

Parameters

Parameter	Meaning
`operation` string · Required	The same operation selected during preview. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`target_version` string · Required	Target version returned by preview. If omitted: No default

See the shared data model.

Request example

JSON Source: example-194.json

```
{
  "operation": "use_sd",
  "active_version": "CURRENT_VERSION",
  "target_version": "PREVIEW_TARGET_VERSION"
}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Selection/recovery can replace the entire hosts/commands/rules/variables/sources set. Preview and preserve current input first.

automation.config.backup

Save the complete emergency configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/config/backup`
Minimum access: ROOT

Parameters

Parameter	Meaning
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-195.json

```
{"active_version":"CURRENT_VERSION"}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

automation.config.enable

Explicitly enable or pause the current configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/config/enable`
Minimum access: ROOT

Parameters

Parameter	Meaning
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`enabled` boolean · Required	true enables the current configuration; false pauses it. If omitted: No default

See the shared data model.

Request example

JSON Source: example-196.json

```
{"active_version":"CURRENT_VERSION","enabled":false}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

automation.config.preview

Preview a configuration restore or source choice

Methods: POST
Path: `/api/v1/automation/config/preview`
Minimum access: ROOT

Parameters

Parameter	Meaning
`operation` string · Required	use_sd/save_to_sd/restore_internal. If omitted: No default

See the shared data model.

Request example

JSON Source: example-197.json

```
{"operation":"use_sd"}
```

Response and completion

Preview differences,current/target versions,and selection preconditions. Apply must use the returned versions.

Usage notes

Selection/recovery can replace the entire hosts/commands/rules/variables/sources set. Preview and preserve current input first.

automation.config.refresh

Check SD card once without changing the user choice

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/config/refresh`
Minimum access: ROOT

Parameters

No business parameters.

See the shared data model.

Request example

JSON Source: example-198.json

```
{}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

automation.config.status

Current automation storage and backup status

Methods: GET / POST
Path: `/api/v1/automation/config/status`
Minimum access: READ

Parameters

No business parameters.

See the shared data model.

Request example

JSON Source: example-199.json

```
{}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

automation.pause

Pause automation engine

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/pause`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-200.json

```
{}
```

Response and completion

code=0/message reports success. Read automation.status to verify runtime state.

Usage notes

Engine state,unfinished actions,or storage can reject requests. A stop timeout does not establish that all actions ended.

automation.proxy.fetch

Proxy fetch from external REST API

Methods: POST
Path: `/api/v1/automation/proxy/fetch`
Minimum access: READ

Parameters

Parameter	Meaning
`url` string · Required	HTTP(S) destination or a supported local API address. If omitted: No default
`method` string · Optional	Supports GET/POST/PUT/DELETE; no generic request-body parameter. If omitted: GET
`headers` object · Optional	String-valued request headers; keep credentials out of ordinary logs. If omitted: None
`timeout` integer · Optional	Total request timeout in milliseconds. If omitted: 15000
`cert_pem` string · Optional	Private-issuer CA material, at most 16384 bytes. If omitted: System trust
`skip_verify` boolean · Optional	Compatibility parameter: false only; true is rejected. If omitted: false

Request example

JSON Source: example-201.json

```
{"url":"https://example.com/status","method":"GET"}
```

Response and completion

data.status/content_length/body; local API results are nested in body, whose code must also be checked.

Usage notes

Local API forwarding retains the current session and destination permissions. External HTTP(S) requests require ROOT. Time, size, and network budgets apply; this is not an unrestricted proxy.

automation.proxy.socketio_test

Test Socket.IO connection (v4 protocol)

Methods: POST
Path: `/api/v1/automation/proxy/socketio_test`
Minimum access: ROOT

Parameters

Parameter	Meaning
`url` string · Required	Socket.IO v4 server address. If omitted: No default
`event` string · Optional	Target event name. If omitted: First event
`timeout_ms` integer · Optional	Connection/event deadline in milliseconds. If omitted: 10000
`json_path` string · Optional	Event-data extraction path. If omitted: Complete data

Request example

JSON Source: example-202.json

```
{"url":"https://example.com","event":"status"}
```

Response and completion

data.connected/sid/event/data, possibly value/path/path_error/error.

Usage notes

Uses Socket.IO rather than plain WebSocket; connects to a real remote service.

automation.proxy.websocket_test

Test WebSocket connection

Methods: POST
Path: `/api/v1/automation/proxy/websocket_test`
Minimum access: ROOT

Parameters

Parameter	Meaning
`uri` string · Required	ws:// or wss:// destination. If omitted: No default
`timeout_ms` integer · Optional	Connection/data wait deadline in milliseconds. If omitted: 10000
`subscribe` string · Optional	Subscription text to send after connecting. If omitted: Send nothing
`json_path` string · Optional	JSON extraction path for the response. If omitted: Complete message

Request example

JSON Source: example-203.json

```
{"uri":"wss://example.com/events","timeout_ms":10000}
```

Response and completion

data.connected/message, possibly value/path/path_error/error.

Usage notes

Connects and can send text. One test is not a persistent subscription.

automation.reload

Reload configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/reload`
Minimum access: ROOT

Parameters

No business parameters.

Request example

JSON Source: example-204.json

```
{}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Engine state,unfinished actions,or storage can reject requests. A stop timeout does not establish that all actions ended.

automation.resume

Resume automation engine

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/resume`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-205.json

```
{}
```

Response and completion

code=0/message reports success. Read automation.status to verify runtime state.

Usage notes

Engine state,unfinished actions,or storage can reject requests. A stop timeout does not establish that all actions ended.

automation.rules.add

Add a new rule

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/rules/add`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`expected_revision` integer · Optional	Original rule revision; carry the latest read value for update. If omitted: Not needed for creation

See the shared data model.

Request example

JSON Source: example-206.json

```
{
  "id": "example-rule",
  "name": "Temperature example",
  "enabled": false,
  "logic": "and",
  "conditions": [
    {
      "variable": "example.temperature",
      "operator": "gt",
      "value": 50
    }
  ],
  "actions": [
    {
      "type": "log",
      "level": 3,
      "message": "Temperature high"
    }
  ],
  "cooldown_ms": 10000,
  "active_version": "CURRENT_VERSION"
}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Saving and triggering are separate. Do not overwrite complete rules from filtered views.

automation.rules.delete

Delete a rule

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/rules/delete`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`preview` boolean · Optional	true previews impact; false performs deletion. If omitted: false
`replacement_id` string · Optional	Replacement object ID for dependent references. If omitted: None
`mode` string · Optional	cascade requests cascade handling; choose one strategy rather than combining force/replacement. If omitted: Ordinary deletion
`force` boolean · Optional	Force deletion of a referenced object only after previewing impact. If omitted: false
`pause_automation` boolean · Optional	Pause automation before handling. If omitted: false

See the shared data model.

Request example

JSON Source: example-207.json

```
{"id":"example-rule","active_version":"CURRENT_VERSION","preview":true}
```

Response and completion

Configuration-result model, possibly impact.

automation.rules.disable

Disable a rule

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/rules/disable`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`expected_revision` integer · Optional	Current rule revision. If omitted: Recommended

See the shared data model.

Request example

JSON Source: example-208.json

```
{"id":"example-rule","active_version":"CURRENT_VERSION"}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

automation.rules.enable

Enable a rule

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/rules/enable`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`expected_revision` integer · Optional	Current rule revision. If omitted: Recommended

See the shared data model.

Request example

JSON Source: example-209.json

```
{"id":"example-rule","active_version":"CURRENT_VERSION"}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

automation.rules.export

Export rule as config pack (.tscfg)

Methods: POST
Path: `/api/v1/automation/rules/export`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`recipient_cert` string · Optional	Recipient certificate PEM; omission encrypts for this device. If omitted: This device certificate

See the shared data model.

Request example

JSON Source: example-210.json

```
{"id":"example-rule"}
```

Response and completion

data.tscfg/filename; package contains rule configuration input.

automation.rules.get

Get rule details by ID

Methods: GET / POST
Path: `/api/v1/automation/rules/get`
Minimum access: READ

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default

See the shared data model.

Request example

JSON Source: example-211.json

```
{"id":"example-rule"}
```

Response and completion

Rule definition,current revision,and runtime statistics; below-ADMIN omits complete conditions/actions.

automation.rules.import

Import rule from config pack

Methods: POST
Path: `/api/v1/automation/rules/import`
Minimum access: ROOT

Parameters

Parameter	Meaning
`tscfg` string · Required	Package JSON text. If omitted: No default
`preview` boolean · Optional	Preview or apply mode. If omitted: false
`active_version` string · Required for apply	Current configuration version. If omitted: No default
`overwrite` boolean · Optional	Overwrite rules with the same ID. If omitted: false
`expected_revision` integer · Required for apply	Rule revision returned by preview. If omitted: No default
`expected_generation` integer · Required for apply	Rule generation returned by preview. If omitted: No default
`credential_generation` integer · Required for apply	Certificate/credential generation returned by preview. If omitted: No default
`package_digest` string · Required for apply	Package digest returned by preview. If omitted: No default

See the shared data model.

Request example

JSON Source: example-212.json

```
{"tscfg":"PACKAGE_JSON_TEXT","preview":true}
```

Response and completion

Preview/apply results,versions,and digests. Apply confirms that preview preconditions still hold.

Usage notes

Complete JSON body only; no query. Confirmation fields differ from other imports; do not use expected_credential/expected_digest here.

automation.rules.list

List all rules

Methods: GET / POST
Path: `/api/v1/automation/rules/list`
Minimum access: READ

Parameters

No business parameters.

See the shared data model.

Request example

JSON Source: example-213.json

```
{}
```

Response and completion

data.rules/count/active_version/loaded/paused/load_error/recovery_required/invalid_items/restart_required. Entries include ID,enablement,revision,runtime statistics; below-ADMIN is restricted.

automation.rules.trigger

Manually trigger a rule

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/rules/trigger`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

Request example

JSON Source: example-214.json

```
{"id":"example-rule","active_version":"CURRENT_VERSION"}
```

Response and completion

Trigger result or rejection reason. Confirm action/service state to establish completion.

Usage notes

Has real effects. Cooldown,pause,conditions,and runtime state may reject triggering.

automation.rules.update

Update rule with expected revision

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/rules/update`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`expected_revision` integer · Optional	Original rule revision; carry the latest read value for update. If omitted: Not needed for creation

See the shared data model.

Request example

JSON Source: example-215.json

```
{
  "id": "example-rule",
  "name": "Temperature example",
  "enabled": false,
  "logic": "and",
  "conditions": [
    {
      "variable": "example.temperature",
      "operator": "gt",
      "value": 50
    }
  ],
  "actions": [
    {
      "type": "log",
      "level": 3,
      "message": "Temperature high"
    }
  ],
  "cooldown_ms": 10000,
  "active_version": "CURRENT_VERSION"
}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Saving and triggering are separate. Do not overwrite complete rules from filtered views.

automation.services.list

List independent remote service controls

Methods: GET / POST
Path: `/api/v1/automation/services/list`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-216.json

```
{}
```

Response and completion

data.items with run_id/command_id/rule_id/source/name/state/current/last_known and operation status.

Usage notes

Retained run records do not prove a remote process still exists. Request verification explicitly when needed.

automation.services.status

Confirm managed remote process state

Methods: GET / POST
Path: `/api/v1/automation/services/status`
Minimum access: READ

Parameters

Parameter	Meaning
`run_id` string · Required	Run ID from the start response or service list. If omitted: No default
`verify` boolean · Optional	Connect to the original host to verify state. If omitted: Per-operation handling

Request example

JSON Source: example-217.json

```
{"run_id":"YOUR_RUN_ID","verify":true}
```

Response and completion

data.run_id/state/source/last_known/confirmed_ms/generation/busy,operation_id/operation_phase/operation_kind/operation_reason/operation_error.

Usage notes

Continue querying busy/in-progress operations. Stop uses verified original process identity, not arbitrary name-based killing.

automation.services.stop

Stop verified managed remote process

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/services/stop`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`run_id` string · Required	Run ID from the start response or service list. If omitted: No default
`verify` boolean · Optional	Connect to the original host to verify state. If omitted: Per-operation handling

Request example

JSON Source: example-218.json

```
{"run_id":"YOUR_RUN_ID","verify":true}
```

Response and completion

data.run_id/state/source/last_known/confirmed_ms/generation/busy,operation_id/operation_phase/operation_kind/operation_reason/operation_error.

Usage notes

Continue querying busy/in-progress operations. Stop uses verified original process identity, not arbitrary name-based killing.

automation.sources.add

Add a new data source

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/sources/add`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-219.json

```
{
  "id": "example-source",
  "label": "Temperature",
  "type": "rest",
  "enabled": false,
  "poll_interval_ms": 5000,
  "rest_url": "https://example.com/status",
  "rest_method": "GET",
  "auto_discover": false,
  "mappings": [
    {
      "path": "temperature",
      "var": "compute.temperature"
    }
  ],
  "active_version": "CURRENT_VERSION"
}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Add requires an unused ID; update requires an existing object. See complete model examples.

automation.sources.add_mapping

Add a field mapping to an existing data source

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/sources/add_mapping`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`json_path` string · Required	Data field path to extract. If omitted: No default
`var_name` string · Required	Destination variable name. If omitted: No default
`transform` string · Optional	Optional transform expression. If omitted: None

See the shared data model.

Request example

JSON Source: example-220.json

```
{
  "id": "example-source",
  "active_version": "CURRENT_VERSION",
  "json_path": "temperature",
  "var_name": "compute.temperature"
}
```

Response and completion

Configuration-result model plus source_id/mapping_count/mapping_index.

automation.sources.delete

Delete a data source

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/sources/delete`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`preview` boolean · Optional	true previews impact; false performs deletion. If omitted: false
`replacement_id` string · Optional	Replacement object ID for dependent references. If omitted: None
`mode` string · Optional	cascade requests cascade handling; choose one strategy rather than combining force/replacement. If omitted: Ordinary deletion
`force` boolean · Optional	Force deletion of a referenced object only after previewing impact. If omitted: false
`pause_automation` boolean · Optional	Pause automation before handling. If omitted: false

See the shared data model.

Request example

JSON Source: example-221.json

```
{"id":"example","active_version":"CURRENT_VERSION","preview":true}
```

Response and completion

Configuration-result model plus impact describing references affected.

Usage notes

Deleting local definitions does not automatically stop remote programs or revoke public keys.

automation.sources.disable

Disable a data source

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/sources/disable`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-222.json

```
{"id":"example-source","active_version":"CURRENT_VERSION"}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

automation.sources.enable

Enable a data source

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/sources/enable`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-223.json

```
{"id":"example-source","active_version":"CURRENT_VERSION"}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

automation.sources.export

Export data source as config pack (.tscfg)

Methods: POST
Path: `/api/v1/automation/sources/export`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`recipient_cert` string · Optional	Recipient certificate PEM; omission encrypts for this device. If omitted: This device certificate

See the shared data model.

Request example

JSON Source: example-224.json

```
{"id":"example"}
```

Response and completion

data.tscfg/filename. Export additionally depends on Developer device identity and recipient materials.

Usage notes

The client saves returned package text. It does not become another device's active configuration automatically.

automation.sources.import

Import data source from config pack

Methods: POST
Path: `/api/v1/automation/sources/import`
Minimum access: ROOT

Parameters

Parameter	Meaning
`tscfg` string · Required	Complete package JSON text supplied as a string. If omitted: No default
`preview` boolean · Optional	Preview only when true. If omitted: false
`active_version` string · Required for apply	Current configuration version. If omitted: Preview may use current version
`overwrite` boolean · Optional	Permit replacing an existing object with different content. If omitted: false
`expected_credential` integer · Required for apply	credential_generation returned by preview. If omitted: None
`expected_digest` string · Required for apply	package_digest returned by preview. If omitted: None

See the shared data model.

Request example

JSON Source: example-225.json

```
{"tscfg":"PACKAGE_JSON_TEXT","preview":true}
```

Response and completion

Configuration-result model. Preview adds valid/items/exists/unchanged/package_digest/credential_generation; apply adds imported/overwritten/restart_required.

Usage notes

Preview first, then apply the same package with digest,credential generation,and configuration version. Preview again if conditions change.

automation.sources.list

List all data sources

Methods: GET / POST
Path: `/api/v1/automation/sources/list`
Minimum access: READ

Parameters

Parameter	Meaning
`offset` integer · Optional	Nonnegative return offset. If omitted: 0
`limit` integer · Optional	Return count; omission does not request truncation. If omitted: All

See the shared data model.

Request example

JSON Source: example-226.json

```
{}
```

Response and completion

data.sources/count/total/active_version/load_error/paused; non-root gets restricted metadata.

automation.sources.remove_mapping

Remove a field mapping from a data source

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/sources/remove_mapping`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`index` integer · One selection alternative	Nonnegative mapping index. If omitted: None
`json_path` string · One selection alternative	Existing field path. If omitted: None
`var_name` string · One selection alternative	Existing variable name. If omitted: None

See the shared data model.

Request example

JSON Source: example-227.json

```
{"id":"example-source","active_version":"CURRENT_VERSION","index":0}
```

Response and completion

Configuration-result model plus source_id/mapping_count.

automation.sources.update

Update an existing data source

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/sources/update`
Minimum access: ROOT

Parameters

Parameter	Meaning
`id` string · Required	Stable object ID; retain the existing ID after reading. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default

See the shared data model.

Request example

JSON Source: example-228.json

```
{
  "id": "example-source",
  "label": "Temperature",
  "type": "rest",
  "enabled": false,
  "poll_interval_ms": 5000,
  "rest_url": "https://example.com/status",
  "rest_method": "GET",
  "auto_discover": false,
  "mappings": [
    {
      "path": "temperature",
      "var": "compute.temperature"
    }
  ],
  "active_version": "CURRENT_VERSION"
}
```

Response and completion

Configuration result with active_version/source/loaded/configured/enabled/paused/saved/uncertain and error location. See the complete config-result model.

Usage notes

Add requires an unused ID; update requires an existing object. See complete model examples.

automation.start

Start automation engine

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/start`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-229.json

```
{}
```

Response and completion

code=0/message reports success. Read automation.status to verify runtime state.

Usage notes

Engine state,unfinished actions,or storage can reject requests. A stop timeout does not establish that all actions ended.

automation.status

Get automation engine status

Methods: GET / POST
Path: `/api/v1/automation/status`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-230.json

```
{}
```

Response and completion

data contains engine state, counters, configuration source, and automatic-start information, depending on engine state.

automation.stop

Stop automation engine

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/stop`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-231.json

```
{}
```

Response and completion

code=0/message reports success. Read automation.status to verify runtime state.

Usage notes

Engine state,unfinished actions,or storage can reject requests. A stop timeout does not establish that all actions ended.

automation.test.device

Test device control action

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/test/device`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Required	agx or lpmu. If omitted: No default
`action` string · Required	A power/reset action supported by the target. If omitted: No default

Request example

JSON Source: example-232.json

```
{"device":"agx","action":"on"}
```

Response and completion

code=0/message reports control-call success. Read device.status afterward.

Usage notes

Controls the real target computer.

automation.test.gpio

Test GPIO action

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/test/gpio`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`pin` integer · Required	Supported pin number. If omitted: No default
`level` boolean · Required	true=high, false=low. If omitted: No default
`pulse_ms` integer · Optional	Pulse duration in milliseconds; 0 means no timed pulse. If omitted: 0

Request example

JSON Source: example-233.json

```
{"pin":1,"level":false,"pulse_ms":0}
```

Response and completion

code=0/message reports action-call success. Read the target pin separately.

Usage notes

Operates hardware pins. Verify their function first.

automation.test.led

Test LED action

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/test/led`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Required	Target LED device. If omitted: No default
`index` integer · Optional	LED index. If omitted: 0
`r` integer · Optional	Red component, 0-255. If omitted: 0
`g` integer · Optional	Green component, 0-255. If omitted: 0
`b` integer · Optional	Blue component, 0-255. If omitted: 0

Request example

JSON Source: example-234.json

```
{"device":"board","r":0,"g":64,"b":128}
```

Response and completion

Action execution result in data.

Usage notes

Despite test in the name, changes the device.

automation.test.ssh

Test SSH command execution

Methods: POST
Path: `/api/v1/automation/test/ssh`
Minimum access: ROOT

Parameters

Parameter	Meaning
`host_id` string · Required	Saved SSH host configuration ID. If omitted: No default
`command` string · Required	Remote command. If omitted: No default
`timeout_ms` integer · Optional	Timeout in milliseconds. If omitted: 30000

Request example

JSON Source: example-235.json

```
{"host_id":"example-host","command":"uname -a","timeout_ms":30000}
```

Response and completion

Remote execution result; inspect exit code and output.

Usage notes

Connects and executes; not a syntax-only check.

automation.variables.delete

Delete a saved user variable

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/variables/delete`
Minimum access: ROOT

Parameters

Parameter	Meaning
`name` string · Required	User-variable name. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`preview` boolean · Optional	Preview reference impact only. If omitted: false
`force` boolean · Optional	Request forced deletion after confirming impact. If omitted: false
`pause_automation` boolean · Optional	Pause automation before processing. If omitted: false

See the shared data model.

Request example

JSON Source: example-236.json

```
{"name":"example.temperature","active_version":"CURRENT_VERSION","preview":true}
```

Response and completion

Configuration-result model, possibly impact.

automation.variables.get

Get variable value

Methods: GET / POST
Path: `/api/v1/automation/variables/get`
Minimum access: READ

Parameters

Parameter	Meaning
`name` string · Required	Complete variable name. If omitted: No default

See the shared data model.

Request example

JSON Source: example-237.json

```
{"name":"compute.temperature"}
```

Response and completion

data contains the variable value and type, source, readonly, and update metadata.

automation.variables.list

List all variables

Methods: GET / POST
Path: `/api/v1/automation/variables/list`
Minimum access: READ

Parameters

Parameter	Meaning
`configured_only` boolean · Optional	List only saved user-variable definitions. If omitted: false
`prefix` string · Optional	Variable-name prefix filter. If omitted: None
`source_id` string · Optional	Filter by source ID. If omitted: None
`include_value` boolean · Optional	Include variable values. If omitted: true
`include_meta` boolean · Optional	Include timestamps, source, readonly, and other metadata. If omitted: true

See the shared data model.

Request example

JSON Source: example-238.json

```
{}
```

Response and completion

data.variables/count. include_value controls value inclusion; include_meta controls time, source, readonly, and related metadata.

automation.variables.set

Set variable value

Methods: POST / PUT / DELETE
Path: `/api/v1/automation/variables/set`
Minimum access: ROOT

Parameters

Parameter	Meaning
`name` string · Required	User-variable name. If omitted: No default
`active_version` string · Required	Latest active_version from configuration state; reread after changes. If omitted: No default
`value` boolean / number / string / null · Required	Value to save; do not substitute arrays/objects for numeric temperatures. If omitted: No default
`create_only` boolean · Optional	Return an existing value without overwriting it. If omitted: false
`source_id` string · Optional	Source identifier. If omitted: user
`value_type` integer · Optional	Optional type ID; normally infer from value. If omitted: Inferred from value

See the shared data model.

Request example

JSON Source: example-239.json

```
{"name":"example.temperature","value":42.5,"active_version":"CURRENT_VERSION"}
```

Response and completion

Configuration-result model plus name/created/value.

Usage notes

Readonly/system-source variables may reject changes. Arrays/objects are outside this simplified value model.

Lights and matrix APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
led.boot.config	Get LED boot configuration
led.brightness	Get/set device brightness
led.clear	Clear all LEDs on device
led.color.hsv	Convert HSV to RGB
led.color.parse	Parse color string to RGB
led.color_correction.export	Export color correction config to SD card
led.color_correction.get	Get color correction configuration
led.color_correction.import	Import color correction config from SD card
led.color_correction.reset	Reset color correction to defaults
led.color_correction.set	Set color correction configuration
led.effect.list	List available effects
led.effect.start	Start effect on device
led.effect.stop	Stop effect on device
led.fill	Fill all LEDs with color
led.filter.list	List available post-processing filters
led.filter.start	Apply post-processing filter
led.filter.stop	Stop post-processing filter
led.image	Display image on matrix
led.list	List LED devices
led.qrcode	Generate and display QR code
led.save	Save current state as boot configuration
led.set	Set an LED pixel range
led.text	Display text on matrix
led.text.stop	Stop text overlay

led.boot.config

Get LED boot configuration

Methods: GET / POST
Path: `/api/v1/led/boot/config`
Minimum access: READ

Parameters

Parameter	Meaning
`device` string · Optional	LED device name: touch/board/matrix; use led.list as authority. If omitted: All LED devices

Request example

JSON Source: example-240.json

```
{"device":"matrix"}
```

Response and completion

data.boot_config array:
device/enabled/animation/speed/brightness/filter/filter_speed/image_path/qrcode_text.

led.brightness

Get/set device brightness

Methods: POST / PUT / DELETE
Path: `/api/v1/led/brightness`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	LED device name: touch/board/matrix; use led.list as authority. If omitted: All LED devices
`brightness` integer · Optional	Brightness, 0-255. If omitted: Query only

Request example

JSON Source: example-241.json

```
{"device":"matrix","brightness":64}
```

Response and completion

data.device/brightness.

led.clear

Clear all LEDs on device

Methods: POST / PUT / DELETE
Path: `/api/v1/led/clear`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	LED device name: touch/board/matrix; use led.list as authority. If omitted: All LED devices
`off` boolean · Optional	Disable output after clearing. If omitted: false

Request example

JSON Source: example-242.json

```
{"device":"matrix"}
```

Response and completion

data.device/cleared.

Usage notes

Stops/replaces the current display. Saving startup configuration is separate.

led.color.hsv

Convert HSV to RGB

Methods: GET / POST
Path: `/api/v1/led/color/hsv`
Minimum access: READ

Parameters

Parameter	Meaning
`h` integer · Required	Integer hue, 0-359 degrees. If omitted: No default
`s` integer · Required	Integer saturation, 0-255. If omitted: No default
`v` integer · Required	Integer value/brightness, 0-255. If omitted: No default

See the shared data model.

Request example

JSON Source: example-243.json

```
{"h":200,"s":255,"v":255}
```

Response and completion

data.hsv/rgb/hex; see the color model for ranges.

led.color.parse

Parse color string to RGB

Methods: GET / POST
Path: `/api/v1/led/color/parse`
Minimum access: READ

Parameters

Parameter	Meaning
`color` string · Required	Supported color string, such as #00A0FF. If omitted: No default

Request example

JSON Source: example-244.json

```
{"color":"#00A0FF"}
```

Response and completion

data.input/rgb(r/g/b)/hex/value.

led.color_correction.export

Export color correction config to SD card

Methods: POST / PUT / DELETE
Path: `/api/v1/led/color_correction/export`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`path` string · Optional	Absolute SD destination path. If omitted: /sdcard/config/led_color_correction.json

Request example

JSON Source: example-245.json

```
{"path":"/sdcard/example-color.json"}
```

Response and completion

data.path/success.

led.color_correction.get

Get color correction configuration

Methods: GET / POST
Path: `/api/v1/led/color_correction/get`
Minimum access: READ

Parameters

No business parameters.

See the shared data model.

Request example

JSON Source: example-246.json

```
{}
```

Response and completion

Color-correction configuration: enabled/brightness/gamma/saturation/white_point; see the color model.

led.color_correction.import

Import color correction config from SD card

Methods: POST / PUT / DELETE
Path: `/api/v1/led/color_correction/import`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`path` string · Required	Absolute device path; managed configuration and sensitive files have additional restrictions. If omitted: No default

Request example

JSON Source: example-247.json

```
{"path":"/sdcard/example-color.json"}
```

Response and completion

data.success.

Usage notes

File reads use the current session. Managed/sensitive paths can require ADMIN.

led.color_correction.reset

Reset color correction to defaults

Methods: POST / PUT / DELETE
Path: `/api/v1/led/color_correction/reset`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-248.json

```
{}
```

Response and completion

data.success.

led.color_correction.set

Set color correction configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/led/color_correction/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`enabled` boolean · Optional	Enable color correction. If omitted: Unchanged
`brightness` object · Optional	{enabled:boolean,factor:number}; factor 0-2. If omitted: Unchanged
`gamma` object · Optional	{enabled:boolean,gamma:number}; gamma 0.1-4; value is a gamma alias. If omitted: Unchanged
`saturation` object · Optional	{enabled:boolean,factor:number}; factor 0-2. If omitted: Unchanged
`white_point` object · Optional	White-point RGB gains; see the color model. If omitted: Unchanged

See the shared data model.

Request example

JSON Source: example-249.json

```
{"enabled":true}
```

Response and completion

data.success.

Usage notes

Read current settings and change only intended fields. Correction differs from per-device brightness 0-255.

led.effect.list

List available effects

Methods: GET / POST
Path: `/api/v1/led/effect/list`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-250.json

```
{}
```

Response and completion

data.effects; led.list also reports device support.

led.effect.start

Start effect on device

Methods: POST / PUT / DELETE
Path: `/api/v1/led/effect/start`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	LED device name: touch/board/matrix; use led.list as authority. If omitted: All LED devices
`effect` string · Required	Effect name from the supported list. If omitted: No default
`speed` integer · Optional	Animation speed, 1-100. If omitted: 50
`color` string / object / integer · Optional	Color string, {r,g,b}, or a decimal 0xRRGGBB integer; components 0-255. If omitted: Effect default color

Request example

JSON Source: example-251.json

```
{"device":"matrix","effect":"rainbow","speed":50}
```

Response and completion

data.device/effect/speed/started.

Usage notes

Confirm the example effect is supported. A new effect replaces the current one.

led.effect.stop

Stop effect on device

Methods: POST / PUT / DELETE
Path: `/api/v1/led/effect/stop`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	LED device name: touch/board/matrix; use led.list as authority. If omitted: All LED devices

Request example

JSON Source: example-252.json

```
{"device":"matrix"}
```

Response and completion

data.device/stopped.

led.fill

Fill all LEDs with color

Methods: POST / PUT / DELETE
Path: `/api/v1/led/fill`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	LED device name: touch/board/matrix; use led.list as authority. If omitted: All LED devices
`color` string / object / integer · Required	Color string, {r,g,b}, or a decimal 0xRRGGBB integer; components 0-255. If omitted: No default

Request example

JSON Source: example-253.json

```
{"device":"board","color":"#00A0FF"}
```

Response and completion

data.device/color(r/g/b)/success.

led.filter.list

List available post-processing filters

Methods: GET / POST
Path: `/api/v1/led/filter/list`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-254.json

```
{}
```

Response and completion

data.filters with name/description.

led.filter.start

Apply post-processing filter

Methods: POST / PUT / DELETE
Path: `/api/v1/led/filter/start`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	LED device name: touch/board/matrix; use led.list as authority. If omitted: All LED devices
`filter` string · Required	Select from led.filter.list. If omitted: No default
`speed` integer · Optional	Filter speed. If omitted: 50
`intensity` number · Optional	Intensity. If omitted: Filter default
`amount` number · Optional	Effect amount. If omitted: Filter default
`scale` number · Optional	Spatial scale. If omitted: Filter default
`angle` number · Optional	Angle. If omitted: Filter default
`frequency` number · Optional	Frequency. If omitted: Filter default
`amplitude` number · Optional	Amplitude. If omitted: Filter default
`wavelength` number · Optional	Wavelength. If omitted: Filter default
`width` number · Optional	Width. If omitted: Filter default
`saturation` number · Optional	Saturation. If omitted: Filter default
`levels` number · Optional	Number of levels. If omitted: Filter default
`decay` number · Optional	Decay. If omitted: Filter default
`density` number · Optional	Density. If omitted: Filter default

Request example

JSON Source: example-255.json

```
{"device":"matrix","filter":"YOUR_FILTER","speed":50}
```

Response and completion

data.device/filter/speed/applied.

Usage notes

Options affect only filters using them. Configure the selected filter rather than applying every option indiscriminately.

led.filter.stop

Stop post-processing filter

Methods: POST / PUT / DELETE
Path: `/api/v1/led/filter/stop`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	LED device name: touch/board/matrix; use led.list as authority. If omitted: All LED devices

Request example

JSON Source: example-256.json

```
{"device":"matrix"}
```

Response and completion

data.device/stopped.

led.image

Display image on matrix

Methods: POST / PUT / DELETE
Path: `/api/v1/led/image`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	Target display device. If omitted: matrix
`path` string · Required	Absolute device path; managed configuration and sensitive files have additional restrictions. If omitted: No default
`center` boolean · Optional	Center the image. If omitted: true

Request example

JSON Source: example-257.json

```
{"device":"matrix","path":"/sdcard/example.png","center":true}
```

Response and completion

data.device/path/displayed/width/height/animated/frames.

Usage notes

Requires supported image format and a valid file. Sensitive reads retain path checks.

led.list

List LED devices

Methods: GET / POST
Path: `/api/v1/led/list`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-258.json

```
{}
```

Response and completion

data.devices with name/count/brightness/layout/effects/current; current has animation/speed/on/color.

led.qrcode

Generate and display QR code

Methods: POST / PUT / DELETE
Path: `/api/v1/led/qrcode`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	Target display device. If omitted: matrix
`text` string · Required	Content to encode. If omitted: No default
`ecc` string · Optional	Error-correction level L/M/Q/H. If omitted: L
`bg_image` string · Optional	Optional absolute background-image path. If omitted: None
`color` string / object / integer · Optional	Color string, {r,g,b}, or a decimal 0xRRGGBB integer; components 0-255. If omitted: Default foreground

Request example

JSON Source: example-259.json

```
{"device":"matrix","text":"https://example.com","ecc":"L"}
```

Response and completion

data.device/text/version/size/capacity/displayed.

Usage notes

Capacity depends on text length,matrix size,and ECC. Background reads have path checks.

led.save

Save current state as boot configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/led/save`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	LED device name: touch/board/matrix; use led.list as authority. If omitted: All LED devices

Request example

JSON Source: example-260.json

```
{"device":"board"}
```

Response and completion

data.device/saved/animation/brightness/speed. Treat it as startup configuration only after saving succeeds.

led.set

Set an LED pixel range

Methods: POST / PUT / DELETE
Path: `/api/v1/led/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Required	LED device name: touch/board/matrix; use led.list as authority. If omitted: No default
`color` string / object / integer · Required	Color string, {r,g,b}, or a decimal 0xRRGGBB integer; components 0-255. If omitted: No default
`index` integer · Optional	Zero-based first pixel; must be below the device pixel count. If omitted: 0
`count` integer · Optional	Positive number of consecutive pixels; the range must fit the device. If omitted: 1 with index; otherwise all pixels

Request example

JSON Source: example-261.json

```
{"device":"board","index":0,"count":1,"color":{"r":0,"g":64,"b":128}}
```

Response and completion

data.device/start/count/color (r/g/b)/success.

Usage notes

Stops the target-layer animation and enables the device. Omitting index/count fills the whole device; index alone writes one pixel. A full fill updates the current solid-color preset. Persistent saving is a separate lighting-configuration step.

led.text

Display text on matrix

Methods: POST / PUT / DELETE
Path: `/api/v1/led/text`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	Target device. If omitted: matrix
`text` string · Required	Display text. If omitted: No default
`font` string · Optional	Supported font name. If omitted: cjk; loads /sdcard/fonts/<font>.fnt
`x` integer · Optional	Starting horizontal coordinate. If omitted: Default position
`y` integer · Optional	Starting vertical coordinate. If omitted: Default position
`align` string · Optional	left/center/right. If omitted: left
`scroll` string · Optional	none/left/right/up/down. If omitted: none
`speed` integer · Optional	Scroll speed. If omitted: 30
`loop` boolean · Optional	Loop the scrolling text. If omitted: false
`invert` boolean · Optional	Invert overlapping output. If omitted: false
`color` string / object / integer · Optional	Color string, {r,g,b}, or a decimal 0xRRGGBB integer; components 0-255. If omitted: White

Request example

JSON Source: example-262.json

```
{"device":"matrix","text":"Ready","scroll":"none"}
```

Response and completion

data.device/text/font/displayed.

Usage notes

Character availability depends on the selected font.

led.text.stop

Stop text overlay

Methods: POST / PUT / DELETE
Path: `/api/v1/led/text/stop`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`device` string · Optional	Target device. If omitted: matrix

Request example

JSON Source: example-263.json

```
{"device":"matrix"}
```

Response and completion

data.device/stopped.

Pin maintenance APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
gpio.info	Get GPIO pin info
gpio.list	List all controllable GPIO pins
gpio.set	Set GPIO level
gpio.toggle	Toggle GPIO level

gpio.info

Get GPIO pin info

Methods: GET / POST
Path: `/api/v1/gpio/info`
Minimum access: READ

Parameters

Parameter	Meaning
`pin` integer · pin or name required	Whitelisted GPIO number from gpio.list. If omitted: None
`name` string · pin or name required	Pin name from gpio.list. If omitted: None

Request example

JSON Source: example-264.json

```
{"name":"AGX_RESET"}
```

Response and completion

data.gpio/name/description/level/default/modified.

gpio.list

List all controllable GPIO pins

Methods: GET / POST
Path: `/api/v1/gpio/list`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-265.json

```
{}
```

Response and completion

data.count/pins; entries gpio/name/description/level/default/modified.

Usage notes

Advanced maintenance with whitelisted pins only, not arbitrary GPIO access.

gpio.set

Set GPIO level

Methods: POST / PUT / DELETE
Path: `/api/v1/gpio/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`pin` integer · pin or name required	Whitelisted GPIO number from gpio.list. If omitted: None
`name` string · pin or name required	Pin name from gpio.list. If omitted: None
`level` integer · Required	0 or 1. If omitted: No default

Request example

JSON Source: example-266.json

```
{"name":"AGX_RESET","level":0}
```

Response and completion

data.gpio/name/level.

Usage notes

Pins can control reset,forced shutdown,or USB routing. Verify physical function first.

gpio.toggle

Toggle GPIO level

Methods: POST / PUT / DELETE
Path: `/api/v1/gpio/toggle`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`pin` integer · pin or name required	Whitelisted GPIO number from gpio.list. If omitted: None
`name` string · pin or name required	Pin name from gpio.list. If omitted: None

Request example

JSON Source: example-267.json

```
{"name":"AGX_RESET"}
```

Response and completion

data.gpio/name/previous/level.

Usage notes

Repeats toggle again. Do not replay blindly after a network timeout.

Application-computer network setup APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
network.lpmu_access.log	Get the execution log for an application computer network access run
network.lpmu_access.start	Start application computer upstream-network access setup
network.lpmu_access.status	Get application computer upstream-network access setup status

network.lpmu_access.log

Get the execution log for an application computer network access run

Methods: GET / POST
Path: `/api/v1/network/lpmu_access/log`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`run_id` integer · Required	Current run number returned by start/status. If omitted: No default
`offset` integer · Required	Nonnegative byte offset; use 0 initially. If omitted: No default

Request example

JSON Source: example-268.json

```
{"run_id":1,"offset":0}
```

Response and completion

data.run_id/output/next_offset/done. Continue at next_offset.

Usage notes

At most 1024 bytes per chunk. Stale run_id is rejected. done means the current log tail was read; still inspect task status.

network.lpmu_access.start

Start application computer upstream-network access setup

Methods: POST / PUT / DELETE
Path: `/api/v1/network/lpmu_access/start`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`sudo_password` string · Optional	Remote sudo password, used for this execution only. If omitted: Empty

Request example

JSON Source: example-269.json

```
{}
```

Response and completion

data.run_id/running/stage. Continue querying status/log.

Usage notes

Fixed target 10.10.99.99:22; requires saved host/key, one task at a time. Changes the remote network.

network.lpmu_access.status

Get application computer upstream-network access setup status

Methods: GET / POST
Path: `/api/v1/network/lpmu_access/status`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`diagnostics` boolean · Optional	Include additional timing/resource diagnostics. If omitted: false

Request example

JSON Source: example-270.json

```
{}
```

Response and completion

data.run_id/running/stage, transferred bytes,exit_code/reason/failure_stage,internet_confirmed/configuration_complete,output_tail/output_truncated. Diagnostics appear only when requested.

Usage notes

Completion requires final-stage/script confirmation. Queuing or script exit 0 alone does not establish configured networking.

General and module configuration APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
config.delete	Delete configuration value
config.get	Get configuration value
config.list	List configuration keys
config.module.list	List all config modules
config.module.reset	Reset module configuration
config.module.save	Save module configuration to NVS
config.module.set	Set module configuration
config.module.show	Show module configuration
config.save	Save configuration to storage
config.set	Set configuration value

config.delete

Delete configuration value

Methods: POST / PUT / DELETE
Path: `/api/v1/config/delete`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`key` string · Required	Complete configuration key name. If omitted: No default

Request example

JSON Source: example-271.json

```
{"key":"example.setting"}
```

Response and completion

data.key/deleted.

config.get

Get configuration value

Methods: GET / POST
Path: `/api/v1/config/get`
Minimum access: Public

Parameters

Parameter	Meaning
`key` string · Required	Complete configuration key name. If omitted: No default

Request example

JSON Source: example-272.json

```
{"key":"system.language"}
```

Response and completion

data.key/type/value. Only key=system.language is anonymous; other keys require ADMIN.

config.list

List configuration keys

Methods: GET / POST
Path: `/api/v1/config/list`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-273.json

```
{}
```

Response and completion

data.items/file_keys/nvs_keys/total_keys, depending on registered configuration.

config.module.list

List all config modules

Methods: GET / POST
Path: `/api/v1/config/module/list`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-274.json

```
{}
```

Response and completion

data.modules with name/version/dirty/nvs_namespace/registered; global sequence information may also appear.

config.module.reset

Reset module configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/config/module/reset`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`module` string · Required	net/dhcp/wifi/nat/led/fan/device/system; see the operation for all support. If omitted: No default
`persist` boolean · Optional	Persist the reset result. If omitted: false

Request example

JSON Source: example-275.json

```
{"module":"system","persist":false}
```

Response and completion

data.module/reset/persisted.

Usage notes

Overwrites the selected module. Read and retain needed settings first.

config.module.save

Save module configuration to NVS

Methods: POST / PUT / DELETE
Path: `/api/v1/config/module/save`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`module` string · Required	net/dhcp/wifi/nat/led/fan/device/system; see the operation for all support. If omitted: No default

Request example

JSON Source: example-276.json

```
{"module":"all"}
```

Response and completion

Single-module result, or all with results/success_count/fail_count. Check each success/error.

config.module.set

Set module configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/config/module/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`module` string · Required	net/dhcp/wifi/nat/led/fan/device/system; see the operation for all support. If omitted: No default
`key` string · Required	Complete configuration key name. If omitted: No default
`value` boolean / number / string · Required	Value matching the module field's type. If omitted: No default

See the shared data model.

Request example

JSON Source: example-277.json

```
{"module":"system","key":"timezone","value":"CST-8"}
```

Response and completion

data.module/key/success/dirty. Persist with config.module.save.

config.module.show

Show module configuration

Methods: GET / POST
Path: `/api/v1/config/module/show`
Minimum access: READ

Parameters

Parameter	Meaning
`module` string · Required	net/dhcp/wifi/nat/led/fan/device/system; see the operation for all support. If omitted: No default

See the shared data model.

Request example

JSON Source: example-278.json

```
{"module":"wifi"}
```

Response and completion

Single module: data.module/config/dirty. all can return grouped modules. Below-ADMIN views omit passwords.

config.save

Save configuration to storage

Methods: POST / PUT / DELETE
Path: `/api/v1/config/save`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-279.json

```
{}
```

Response and completion

data.saved. Preserve/query state after failure; do not assume complete persistence.

config.set

Set configuration value

Methods: POST / PUT / DELETE
Path: `/api/v1/config/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`key` string · Required	Complete configuration key name. If omitted: No default
`value` boolean / number / string · Required	Configuration value; not an object or array. If omitted: No default

Request example

JSON Source: example-280.json

```
{"key":"system.language","value":"zh-CN"}
```

Response and completion

data.key/success. Use config.save separately for persistence.

Configuration packages APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
config.pack.apply	Unsupported generic configuration application
config.pack.content	Decrypt and get content of imported config pack
config.pack.export	Export configuration as encrypted .tscfg
config.pack.export_cert	Export device certificate for encryption
config.pack.import	Validate an uploaded .tscfg file in place
config.pack.info	Get config pack system information
config.pack.list	List .tscfg files in directory
config.pack.verify	Verify a .tscfg package signature

config.pack.apply

Unsupported generic configuration application

Methods: POST / PUT / DELETE
Path: `/api/v1/config/pack/apply`
Minimum access: ADMIN

Unsupported operation. The name is retained to explicitly reject obsolete calls.

Parameters

Parameter	Meaning
`path` string · Required	Absolute device path; managed configuration and sensitive files have additional restrictions. If omitted: No default

Request example

JSON Source: example-281.json

```
{"path":"/sdcard/example.tscfg"}
```

Response and completion

data.success=false with unsupported result_code/reason. No configuration is applied even if top-level code=0.

Usage notes

Generic apply is unsupported. Use the object's dedicated import flow.

config.pack.content

Decrypt and get content of imported config pack

Methods: POST / PUT / DELETE
Path: `/api/v1/config/pack/content`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`name` string · Required	Package name as used by package storage. If omitted: No default

Request example

JSON Source: example-282.json

```
{"name":"example"}
```

Response and completion

data.name/content/content_raw; content is structured when parseable JSON.

Usage notes

Trusted administrative reads only. Keep configuration secrets out of public logs.

config.pack.export

Export configuration as encrypted .tscfg

Methods: POST / PUT / DELETE
Path: `/api/v1/config/pack/export`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`name` string · Required	Package name. If omitted: No default
`content` object / string · Required	Configuration content to encrypt; must be JSON-encodable. If omitted: No default
`recipient_cert` string · Required	Recipient device certificate PEM. If omitted: No default
`description` string · Optional	Package description. If omitted: Empty
`save_path` string · Optional	Optional SD save path. If omitted: Do not save a device file

Request example

JSON Source: example-283.json

```
{"name":"example","content":{"example":true},"recipient_cert":"RECIPIENT_CERTIFICATE_PEM"}
```

Response and completion

data.tscfg/filename/size; successful requested file save also includes saved_path.

Usage notes

Requires local Developer export capability. The destination still validates and explicitly imports the package.

config.pack.export_cert

Export device certificate for encryption

Methods: GET / POST
Path: `/api/v1/config/pack/export_cert`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-284.json

```
{}
```

Response and completion

data.certificate/fingerprint/cn; public material for encrypting to this device, not a private key.

config.pack.import

Validate an uploaded .tscfg file in place

Methods: POST / PUT / DELETE
Path: `/api/v1/config/pack/import`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`content` string · content or path required	Raw package JSON text, not an arbitrary Base64 payload. If omitted: No default
`path` string · content or path required	Absolute path of an uploaded package on the device. If omitted: No default

Request example

JSON Source: example-285.json

```
{"path":"/sdcard/example.tscfg"}
```

Response and completion

Success data:valid/result_code/result_message/reason/path/saved_path,name/description/source_file/target_device/created_at,signature. Invalid packages return a business error.

Usage notes

content can save a verified package to package storage; path validates in place. Neither applies business configuration. Use the relevant object's import.

config.pack.info

Get config pack system information

Methods: GET / POST
Path: `/api/v1/config/pack/info`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-286.json

```
{}
```

Response and completion

data.can_export/device_type/cert_fingerprint/cert_cn/cert_ou/pack_version/algorithm.

config.pack.list

List .tscfg files in directory

Methods: GET / POST
Path: `/api/v1/config/pack/list`
Minimum access: READ

Parameters

Parameter	Meaning
`path` string · Optional	Absolute directory path. If omitted: /sdcard/config

Request example

JSON Source: example-287.json

```
{"path":"/sdcard/config"}
```

Response and completion

data.files/count; entries name/path/size/mtime/valid and available issuer information.

config.pack.verify

Verify a .tscfg package signature

Methods: POST / PUT / DELETE
Path: `/api/v1/config/pack/verify`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`content` string · content or path required	Raw package JSON text, not an arbitrary Base64 payload. If omitted: No default
`path` string · content or path required	Absolute path of an uploaded package on the device. If omitted: No default

Request example

JSON Source: example-288.json

```
{"path":"/sdcard/example.tscfg"}
```

Response and completion

data.valid/result_code/result_message/reason/signature. Signature has valid/is_official/signer_cn/signer_ou/signed_at.

Usage notes

Check nested valid/reason. Generic validation does not establish configuration import.

Updates and recovery APIs

Complete index | Common rules | Troubleshooting

Examples are POST JSON bodies with illustrative values. Response descriptions refer to data unless explicitly stated otherwise.

API	Purpose
ota.abort	Request firmware update cancellation
ota.partitions	Inspect update partitions
ota.progress	Read firmware update progress
ota.rollback	Roll back firmware
ota.server.get	Read update-server configuration
ota.server.set	Set update-server configuration
ota.start_file	Start firmware update from SD
ota.start_url	Start firmware update from a URL
ota.status	Read firmware update status
ota.test_connection	Probe an update URL
ota.upgrade_file	Compatibility alias for ota.start_file
ota.upgrade_url	Compatibility alias for ota.start_url
ota.upload_abort	Abort firmware upload control
ota.upload_begin	Begin firmware upload control
ota.upload_end	Finish firmware upload control
ota.validate	Confirm healthy running firmware
ota.version	Read firmware version
ota.www.abort	Request WWW update cancellation
ota.www.progress	Read WWW update progress
ota.www.start	Start WWW update from a URL
ota.www.start_sdcard	Start WWW update from SD

ota.abort

Request firmware update cancellation

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/abort`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-289.json

```
{}
```

Response and completion

data.aborted=true; read the corresponding progress to confirm state.

Usage notes

Acts on the current update, not a private task. Do not immediately start another update while state is unclear.

ota.partitions

Inspect update partitions

Methods: GET / POST
Path: `/api/v1/ota/partitions`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-290.json

```
{}
```

Response and completion

data.running/next each contain label/address/size/is_bootable; bootable partitions include version/project. Also includes can_rollback.

ota.progress

Read firmware update progress

Methods: GET / POST
Path: `/api/v1/ota/progress`
Minimum access: READ

Parameters

No business parameters.

See the shared data model.

Request example

JSON Source: example-291.json

```
{}
```

Response and completion

data.state/percent/total_size/received_size/message. state is idle/checking/downloading/verifying/writing/pending_reboot/error.

Usage notes

This is device-wide progress, not a per-login task result. Poll after starting. 100 percent alone does not prove the device is running the new version.

ota.rollback

Roll back firmware

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/rollback`
Minimum access: ADMIN

Parameters

No business parameters.

See the shared data model.

Request example

JSON Source: example-292.json

```
{}
```

Response and completion

Attempts to select the rollback partition and restart; the connection may close before a response. If returned, data.rolling_back=true.

Usage notes

First check ota.partitions.can_rollback. Log in again after rollback and verify version/configuration compatibility. no_reboot is not supported.

ota.server.get

Read update-server configuration

Methods: GET / POST
Path: `/api/v1/ota/server/get`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-293.json

```
{}
```

Response and completion

ADMIN can read data.url/configured; lower roles receive configured only.

ota.server.set

Set update-server configuration

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/server/set`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`url` string · Required	Update-server URL, at most 255 characters; an empty string clears it. If omitted: No default
`save` boolean · Optional	Persist the server address. If omitted: false

Request example

JSON Source: example-294.json

```
{"url":"http://updates.example.com/firmware.bin","save":true}
```

Response and completion

data.url/saved.

Usage notes

Setting the address does not start an update. The WWW URL can be derived from this address.

ota.start_file

Start firmware update from SD

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/start_file`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`file` string · file or path required	Absolute firmware file path; prefer file. If omitted: No default
`path` string · Optional	Alias of file. If omitted: Not supplied
`auto_reboot` boolean · Optional	Restart automatically after firmware completion. If omitted: true
`no_reboot` boolean · Optional	Inverse alias of auto_reboot; overrides it when supplied. If omitted: Do not override
`allow_downgrade` boolean · Optional	Allow an older version; image validation still applies. If omitted: false
`skip_verify` boolean · Optional	Compatibility option for SD files. There is no TLS connection; it does not bypass firmware completeness checks. If omitted: false

See the shared data model.

Request example

JSON Source: example-295.json

```
{"file":"/sdcard/firmware.bin","no_reboot":true}
```

Response and completion

data.started/path.

Usage notes

Start success means the request began. Read ota.progress and verify ota.status after restart. Use a complete image for this hardware and partition layout.

ota.start_url

Start firmware update from a URL

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/start_url`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`url` string · Required	HTTP or HTTPS URL of the complete firmware image, at most 2048 bytes. If omitted: No default
`auto_reboot` boolean · Optional	Restart automatically after firmware completion. If omitted: true
`no_reboot` boolean · Optional	Inverse alias of auto_reboot; overrides it when supplied. If omitted: Do not override
`allow_downgrade` boolean · Optional	Allow an older version; image validation still applies. If omitted: false
`skip_verify` boolean · Optional	Requests relaxed HTTPS server checks, subject to firmware TLS settings; the connection may still fail. Image completeness checks remain. Normally keep false. If omitted: false

See the shared data model.

Request example

JSON Source: example-296.json

```
{"url":"http://updates.example.com/firmware.bin","no_reboot":true}
```

Response and completion

data.started/url.

Usage notes

Start success means the request began. Read ota.progress and verify ota.status after restart. Use a complete image for this hardware and partition layout.

ota.status

Read firmware update status

Methods: GET / POST
Path: `/api/v1/ota/status`
Minimum access: READ

Parameters

No business parameters.

Request example

JSON Source: example-297.json

```
{}
```

Response and completion

data.state/running/next/pending_verify; rollback_timeout when verification is pending. Partitions include label/address/size and available version information.

Usage notes

size is bytes. After writing, restart and confirm the running version.

ota.test_connection

Probe an update URL

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/test_connection`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`url` string · Required	Update-file URL to probe. If omitted: No default

Request example

JSON Source: example-298.json

```
{"url":"http://updates.example.com/firmware.bin"}
```

Response and completion

data contains success and connection diagnostics.

Usage notes

Makes a real network request. Connectivity does not establish image installability.

ota.upgrade_file

Compatibility alias for ota.start_file

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/upgrade_file`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`file` string · file or path required	Absolute firmware file path; prefer file. If omitted: No default
`path` string · Optional	Alias of file. If omitted: Not supplied
`auto_reboot` boolean · Optional	Restart automatically after firmware completion. If omitted: true
`no_reboot` boolean · Optional	Inverse alias of auto_reboot; overrides it when supplied. If omitted: Do not override
`allow_downgrade` boolean · Optional	Allow an older version; image validation still applies. If omitted: false
`skip_verify` boolean · Optional	Compatibility option for SD files. There is no TLS connection; it does not bypass firmware completeness checks. If omitted: false

See the shared data model.

Request example

JSON Source: example-299.json

```
{"file":"/sdcard/firmware.bin","no_reboot":true}
```

Response and completion

data.started/path.

Usage notes

Compatibility alias with the parameters, permissions, and results of ota.start_file.

ota.upgrade_url

Compatibility alias for ota.start_url

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/upgrade_url`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`url` string · Required	HTTP or HTTPS URL of the complete firmware image, at most 2048 bytes. If omitted: No default
`auto_reboot` boolean · Optional	Restart automatically after firmware completion. If omitted: true
`no_reboot` boolean · Optional	Inverse alias of auto_reboot; overrides it when supplied. If omitted: Do not override
`allow_downgrade` boolean · Optional	Allow an older version; image validation still applies. If omitted: false
`skip_verify` boolean · Optional	Requests relaxed HTTPS server checks, subject to firmware TLS settings; the connection may still fail. Image completeness checks remain. Normally keep false. If omitted: false

See the shared data model.

Request example

JSON Source: example-300.json

```
{"url":"http://updates.example.com/firmware.bin","no_reboot":true}
```

Response and completion

data.started/url.

Usage notes

Compatibility alias with the parameters, permissions, and results of ota.start_url.

ota.upload_abort

Abort firmware upload control

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/upload_abort`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-301.json

```
{}
```

Response and completion

data.aborted.

ota.upload_begin

Begin firmware upload control

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/upload_begin`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`size` integer · Required	Positive total firmware byte count, also limited by the target partition. If omitted: No default

Request example

JSON Source: example-302.json

```
{"size":1048576}
```

Response and completion

data.ready/expected_size.

Usage notes

This controls an upload. There is no public upload_write API for data, so begin/end alone cannot transfer firmware. Ordinary integrations should use the complete-file binary route.

ota.upload_end

Finish firmware upload control

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/upload_end`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`auto_reboot` boolean · Optional	Restart automatically on completion. If omitted: true

Request example

JSON Source: example-303.json

```
{"auto_reboot":false}
```

Response and completion

data.completed/reboot_pending.

Usage notes

Requires an active upload with complete data. Fails when no upload is active.

ota.validate

Confirm healthy running firmware

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/validate`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-304.json

```
{}
```

Response and completion

data.validated.

Usage notes

Confirmation changes rollback state. First confirm the running version and application health.

ota.version

Read firmware version

Methods: GET / POST
Path: `/api/v1/ota/version`
Minimum access: Public

Parameters

No business parameters.

Request example

JSON Source: example-305.json

```
{}
```

Response and completion

data.version/project/compile_date/compile_time/idf_version.

ota.www.abort

Request WWW update cancellation

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/www/abort`
Minimum access: ADMIN

Parameters

No business parameters.

Request example

JSON Source: example-306.json

```
{}
```

Response and completion

data.aborted=true; read the corresponding progress to confirm state.

Usage notes

Acts on the current update, not a private task. Do not immediately start another update while state is unclear.

ota.www.progress

Read WWW update progress

Methods: GET / POST
Path: `/api/v1/ota/www/progress`
Minimum access: READ

Parameters

No business parameters.

See the shared data model.

Request example

JSON Source: example-307.json

```
{}
```

Response and completion

data.state/percent/total/received/message/running. Completion uses state=completed and running=false.

Usage notes

This is device-wide progress, not a per-login task result. Poll after starting. 100 percent alone does not prove the device is running the new version.

ota.www.start

Start WWW update from a URL

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/www/start`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`url` string · Optional	HTTP or HTTPS URL of www.bin. Limit the complete URL to 255 bytes; longer input may not be retained in full. If omitted: Derived from configured update server
`skip_verify` boolean · Optional	Requests relaxed HTTPS server checks, subject to firmware TLS settings; the connection may still fail. Image completeness checks remain. Normally keep false. If omitted: false

See the shared data model.

Request example

JSON Source: example-308.json

```
{"url":"http://updates.example.com/www.bin"}
```

Response and completion

data.status=started/url.

Usage notes

Use the complete www.bin matching the firmware, not an individual HTML file. Then read ota.www.progress.

ota.www.start_sdcard

Start WWW update from SD

Methods: POST / PUT / DELETE
Path: `/api/v1/ota/www/start_sdcard`
Minimum access: ADMIN

Parameters

Parameter	Meaning
`file` string · Required	Absolute file path of the WWW image. Limit it to 255 bytes; longer input may not be retained in full. If omitted: No default

See the shared data model.

Request example

JSON Source: example-309.json

```
{"file":"/sdcard/www.bin"}
```

Response and completion

data.status=started/file.